

Structural Principle of Entanglement (PSE)

*Foundations of Emergence:
physical quantities as images of a fundamental pregeometric structure*

Conceptual Framework

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Abstract

The Structural Principle of Entanglement (PSE) proposes an emergentist framework in which no physical quantity is fundamental. Spacetime, dynamics, fields, statistical distributions, and all physical observables emerge from a real prespatial structure S , devoid of coordinates, metric, causality, and physical properties. The fundamental reorganization $\mathfrak{D}$ of the prespatial structure S generates the discrete fundamental structural distance $D_S^{(f)}$ and the selective binary correlation C_f , regulated by the ontological threshold Δ . The same fundamental reorganization $\mathfrak{D}$, continuing its action on the fundamental structure S , produces the continuous emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e , which constitute a continuous but non-observable pregeometric domain. It is the action of $\mathfrak{D}$ that generates the increase in complexity of the emergent structure: when such complexity exceeds a threshold, the emergent configurations become representable as physical quantities. The map Φ , monotonic, non-invertible, and compressive, precisely formalizes that level of complexity at which the emergent structure becomes physically representable:

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{physical}}.$$

Distinct classes of maps Φ_α formalize different physical domains (geometric, dynamical, statistical, field-theoretic, or discrete), all derived from the same structural information. The informational rigidity of Φ prevents physical quantities from amplifying the emergent correlation, ensuring internal coherence between local and non-local phenomena. Physical constants emerge as compressed images of structural constraints: the threshold Δ and discreteness imply a minimal correlation scale ($\hbar$), the structural coherence limit c becomes the relativistic limit, while the compressivity of Φ determines a response coefficient that in the physical domain takes the form of the gravitational constant G .

Emergence follows a unidirectional and non-invertible ontological chain:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \xrightarrow{\Theta_k} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

Physical observables are effective representations of the emergent correlation: none can reconstruct or influence the fundamental structure. Local and global phenomenology depends on structural variations $\Delta D_S^{(e)}$: local variations generate domains compatible with experimental physics, while global variations produce effective large-scale structures, including the geometric and dynamical representations used in cosmology.

The PSE provides a unified description of the emergence of physical quantities, grounded in the compressivity of the map Φ and consistent with the main contemporary lines of research on pregeometry, information, and the emergence of spacetime.

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Part I - General Introduction

1 Why the name “Structural Principle of Entanglement”

The term *Structural Principle of Entanglement* (PSE) derives from the ontological nature of entanglement within the PSE: it is not a physical property, nor a quantum phenomenon, but a structural consequence of the emergent correlation that forms along the ontological chain of the fundamental domain. Entanglement is not interpreted as an interaction, a dynamics, or a physical process, but rather as a structural constraint imposed by the emergent correlation C_e and the emergent structural distance $D_S^{(e)}$.

The principle is “structural” because it concerns exclusively the pregeometric structure and its regularizations, independently of any physical representation; it is “of entanglement” because the emergent correlation it describes coincides, in the physical domain, with what is observed as entanglement.

The name therefore formalizes the fact that entanglement is a compressed image of a non-physical ontological correlation, and that its origin lies in the emergent structure, not in the dynamics of the physical domain.

2. Reality in the PSE

In the PSE, S is the primitive ontological domain: a real, prespatial level devoid of any physical structure. It is composed of pure relational entities, elements that possess no intrinsic properties and no geometric or dynamical characteristics. A pure relational entity exists only insofar as it is part of a network of relations with other entities of S ; it has no internal identity, no state, no form.

What can change are not the entities themselves, but the relations that connect them. The fundamental reorganization $\mathfrak{D}$ acts exclusively on the relations, generating different configurations of S . From these configurations emerges the fundamental structural distance $D_S^{(f)}$, the first organizational level that prepares the formation of the emergent structure.

2.1 Reality as an Emergent Structure

The fundamental reality described by the PSE does not coincide with the physical domain, nor with the set of measurable quantities. It is constituted by the **emergent structure**, that is, by the complex configuration of correlations that arises from the configuration of the prespatial structure S . This emergent structure, represented by $D_S^{(e)}$ and C_e and defined on the emergent domain X , is real in the strongest sense: it exists autonomously, independently of any observer and of any physical interpretation. Reality is not defined by physical quantities, but by the coherence and stability of the emergent relations that constitute its very nature.

2.2 Emergent Complexity, Discrete World, Macroscopic Objects, and Cosmological Structures

As complexity increases, the emergent structure generates the discrete world, namely the level at which emergent correlations acquire stable, distinguishable, and persistent configurations. The discrete world is not yet a physical domain, but constitutes the first structured form of reality: a set of coherent configurations that manifest as pre-physical discrete units. The discrete units of the discrete world are not entities nor objects: they are coherent configurations of the emergent correlation. They possess no mass, charge, spin, position, or any physical property; they are not localized and do not

exist as individual elements. The discrete world is a structural level, not a collection of objects: its units are patterns of coherence, not physical components. At this level, elementary configurations appear that physics will interpret as “particles” only after the quantum compression performed by Φ_{quant} ; within the discrete world they are not physical entities, but simple structural units.

Macroscopic objects are not aggregates of physical entities, nor collections of particles or fields. In the PSE, they are complex and stable configurations of the discrete world: emergent patterns that maintain coherence through the structural index t , understood as a simple ordering of fundamental configurations. The elementary configurations of the discrete world are not physical particles: they are pre-physical structural units, lacking mass, spin, charge, or any quantum property. Only after the quantum compression performed by Φ_{quant} can these configurations be represented as particles within the physical domain.

Macroscopic objects are more complex extensions of the elementary configurations of the discrete world: stable forms of the emergent correlation that maintain rigidity and coherence on larger scales. Their existence does not depend on physical quantities, but on the structural stability of the emergent correlation. In this sense, macroscopic objects are real but not physical: they belong to emergent reality, not to its physical representation.

The same gradient of complexity that generates discrete units and macroscopic objects continues on increasingly larger scales, producing global configurations of the emergent correlation. When such configurations achieve large-scale coherence, the map Φ represents them as geometric and dynamical structures: curvature, effective metrics, and collective behaviors that physics interprets as cosmological phenomena. The global properties of the Universe (expansion, rigidity, distribution of structures) are therefore compressed images of structural variations in the emergent domain, not fundamental dynamics. Cosmology thus represents the macroscopic form of the same emergent reality that, at lower complexity, produces elementary configurations and objects.

2.3 Reality Is Observable, but Only Beyond a Complexity Threshold

The emergent structure, in its elementary forms, is non-observable in both physical and experiential senses: it produces neither measurable quantities nor perceivable forms. However, when the fundamental reorganization generates sufficiently high levels of relational complexity, the emergent structure becomes accessible to perception. What we see, touch, and recognize is not physics, but the emergent structure at a level of complexity capable of producing objects, forms, and stable regularities.

The essential distinction is therefore threefold:

- **non-observable physically**, because it lacks observables;
- **non-observable experientially at elementary levels**, because it lacks form;
- **observable experientially at complex levels**, where the emergent structure produces perceivable configurations.

The map Φ formalizes what emergent complexity already makes perceivable.

2.4 Reality as an Autonomous and Pre-Physical Domain

Reality in the PSE is an autonomous ontological domain, which does not depend on physics nor on its categories. It exists prior to physics, independently of any quantitative representation. Physics is not reality, but a language that interprets it. The emergent structure constitutes nature; physics is its

translation. This distinction is fundamental for understanding the role of the map Φ and the non-physical nature of the fundamental levels of the PSE.

2.5 Why Reality Must Be Separated from Physics

Traditional physics tends to identify reality with what is measurable. The PSE overturns this perspective: reality is what spontaneously emerges from the configurations of the structure S , whereas physics is an effective representation obtained through Φ . Without this distinction, the physical reader would interpret the emergent structure as “pre-physical” or “proto-spatial,” completely misunderstanding the nature of the theory. The conceptual separation between reality and physics is therefore necessary to preserve the ontological coherence of the PSE.

This distinction makes it possible to understand why the PSE is not a physical theory, but a theory of reality: physics does not coincide with reality, but constitutes a compressed, monotonic, and non-invertible translation of it. Having clarified that the reality of the PSE is constituted by the emergent structure and that physics represents only an effective image of it, we can now introduce the conceptual framework that formalizes this reality in its fundamental elements.

3. Distinction Between Reality and Physics

3.1 Reality and Physics as Distinct Ontological Domains

In the PSE, reality and physics belong to different ontological domains. Reality is constituted by the emergent structure, that is, by the complex configuration of correlations C_e and emergent distances $D_S^{(e)}$ that derive from the structure S . This structure is autonomous, spontaneous, and independent of any interpretation. Physics, by contrast, is not an ontological domain: it is a representational language that describes reality through measurable quantities. The distinction between these two levels is essential for understanding the nature of the PSE.

3.2 Physics as an Effective Representation of Reality

Physics does not coincide with reality, but constitutes an effective representation of it. Physical quantities (distances, times, fields, metrics) do not exist in the emergent domain: they appear only after the application of the map Φ , which interprets the emergent structure as a physical domain. Physics is therefore a system of description, not a fundamental level of being. It translates emergent reality into measurable quantities, but does not generate or modify it.

3.3 The Map Φ as a Bridge Between Reality and Physics

The map Φ is the conceptual element that connects emergent reality to the physical domain. It is not real, not a dynamics, and not a physical process: it is an interpretative transformation that makes physically readable what is already real. Φ does not add information, does not introduce new quantities, and does not alter the emergent structure. Its function is exclusively representational: it allows reality to be expressed in physical terms without modifying its ontological nature.

3.4 Why Reality Is Deeper Than Physics

Emergent reality is deeper than physics because it exists independently of any representation. Physics depends on the map Φ , whereas reality does not. The emergent structure is autonomous, spontaneous, and complete: it does not require observables, metrics, or fields in order to exist. Physics, instead, requires a translation of reality into measurable quantities. In this sense, physics is a derived level, whereas reality is the fundamental level.

3.5 Conceptual Consequences for the Interpretation of the PSE

The distinction between reality and physics prevents the PSE from being interpreted as a traditional physical theory. The emergent structure is not a “pre-space,” nor a “proto-field,” nor a hidden physical domain: it is reality itself. The discrete world is not a set of physical entities, but a complex level of emergent reality. Macroscopic objects are not aggregates of particles, but stable configurations of emergent correlation. Physics describes these objects only after the projection Φ , but does not create them.

3.6 Necessity of the Conceptual Separation

Without a clear separation between reality and physics, the physical reader would interpret the emergent structure through physical categories, misunderstanding the nature of the PSE. The distinction is therefore necessary to preserve the coherence of the theory: reality emerges spontaneously from the fundamental reorganization $\mathfrak{D}$ on S , whereas physics is an effective representation obtained through the projection map Φ .

This separation makes it possible to understand why the PSE is not a physical theory, but a theory of reality: physics does not coincide with reality, but constitutes a compressed, monotonic, and non-invertible translation of it. Having clarified that the reality of the PSE is constituted by the emergent structure and that physics represents only an effective image of it, we can now introduce the conceptual framework that formalizes this reality in its fundamental elements.

Part II - Conceptual Framework

Contemporary physics is described by extremely effective but conceptually incompatible theories, because they assume geometric, dynamical, or statistical structures as primitive. General relativity presupposes a continuous and metric spacetime, whereas quantum mechanics describes discrete, non-local phenomena lacking intrinsic geometry. This tension arises from the idea that observable physical structures are fundamental.

The Structural Principle of Entanglement (PSE) rejects this assumption: no physical quantity is fundamental. All observables emerge from a deeper prespatial structure, S , devoid of coordinates, metric, causality, dynamics, or physical properties. The elements of S are defined only by their structural relations, described by the fundamental structural distance $D_S^{(f)}$. This choice is motivated by the epistemic need to describe non-local correlations without assuming underlying geometry or dynamics.

Through the fundamental structural reorganization $\mathfrak{D}$ on S , the discrete fundamental structural distance $D_S^{(f)}$ generates the selective binary correlation C_f , regulated by the threshold Δ . The collective aggregation of fundamental correlations produces a continuous emergent structural distance $D_S^{(e)}$, belonging to a non-observable but already regular domain. From $D_S^{(e)}$ derives the continuous emergent correlation C_e which represents the pregeometric form of coherence and constitutes the level from which all physical quantities will arise. C_e does not live in spacetime: its nonlocality is a structural, not dynamical, property, and allows non-local physical correlations to be understood without introducing physical transport or superluminal mechanisms.

The fundamental level is not physicalist: it contains no physical quantities, possesses no observable dynamics, and cannot be described using geometric, dynamical, or statistical concepts. Physics emerges only as an effective representation of the successive structural levels.

Physical quantities emerge through a monotonic map

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{physical}}$$

which projects the emergent structure into a physical domain. The same structure $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$ generates distinct physical domains through different classes of maps Φ_α . Each class produces an effective representation (geometric, dynamical, field-theoretic, statistical, or discrete), ensuring that all physical structures emerge from the same structural information. Multi-emergence guarantees coherence among the different physical domains and prevents conflicts between geometry, dynamics, fields, and quantization. The map Φ is compressive: it reduces structural information to a physical representation, which is why no emergent quantity can reconstruct the prespatial structure.

The ontological hierarchy is:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}} \xrightarrow{\Theta_k} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

and is regulated by the fundamental reorganization $\mathfrak{D}$, which incorporates the structural index t , the structural coherence limit c , and the principle $E + C$, and by the threshold Δ , which selects the fundamental correlations that contribute to C_f .

From these constraints physical constants also emerge: discreteness and the threshold Δ produce a minimal scale that appears as the Planck constant; the structural limit c becomes the relativistic limit; the compressivity of Φ generates a response coefficient that in the physical domain takes the form of the gravitational constant G . None of these constants is fundamental: all are compressed images of pregeometric constraints.

The hierarchy is non-retroactive: each level emerges from the previous one without influencing it, and no physical quantity can reconstruct the prespatial structure from which it derives.

The PSE shows that observable physics is not a set of primitive axioms, but the compressed manifestation of a real prespatial structure. The fundamental reorganization $\mathfrak{D}$ acts exclusively on S : it orders and reorganizes the configurations of the prespatial structure, from which all physical quantities emerge as transformations of the same structural information. The PSE does not modify existing theories: it clarifies their ontological origin, showing that geometry, dynamics, causality, and quantum correlations are emergent representations of the same prespatial structure, obtained through the compressive map Φ .

Observability is not a fundamental property of reality, but a characteristic that emerges when the emergent structure reaches a level of complexity sufficient to produce stable and perceivable forms. Emergent reality exists independently of the map: it is real, structured, and, at elementary levels, non-observable.

The map Φ formalizes the portion of the emergent structure that, thanks to its complexity, can be represented as measurable physical quantities in the physical domain.

Technical Note

The tension between the geometric structures of general relativity and the non-local correlations of quantum mechanics is widely recognized in contemporary literature. General relativity assumes a continuous and metric spacetime (Misner, Thorne & Wheeler 1973), whereas quantum mechanics describes discrete, non-local phenomena lacking intrinsic geometry (Bell 1964; Brunner et al. 2014). This incompatibility has motivated approaches that question the fundamentality of observable physical structures. In causal sets (Bombelli et al. 1987; Sorkin 2003), geometric structures emerge from discrete relations. In non-commutative geometry (Connes 1994; Connes & Marcolli 2008), the fundamental structure is algebraic, and physical quantities emerge from spectral properties. In informational models (Van Raamsdonk 2010; Swingle 2012), physical structures are interpreted as images of more primitive correlations. In this context, the PSE's idea that all observables emerge from a prespatial structure S is fully consistent with contemporary research directions.

1. Ontological Foundations of the PSE

1.1 Real, Observable, and Non-Observable

In the PSE, the term **real** denotes an ontologically existing entity, independently of whether it can be measured within the physical domain. *Real* does not coincide with *observable*: a structure may be real while possessing no physical quantity.

An entity is **observable** when it can be measured or inferred within the physical domain, i.e., within the domain generated by the monotonic projection Φ .

An entity is **non-observable** when it possesses no direct physical interaction, while still contributing to the structure of reality through ontological correlations. The following are non-observable:

- the fundamental prespatial structure S ,

- the fundamental structural distance $D_S^{(f)}$,
- the emergent structural distance $D_S^{(e)}$,
- the emergent correlation C_e ,
- the emergent domain X ,
- the emergent structure E .

The physical domain is the only observable level: it arises from the formalization of the emergent structure through Φ .

Technical Note

The distinction between real and observable is consistent with ontological and informational approaches in contemporary physics. In noncommutative geometry (Connes 1994; Connes & Marcolli 2008), there exist real entities that are not directly observable, whose structure determines emergent physical properties. In causal set theory (Bombelli et al. 1987; Sorkin 2003; Dowker 2020), the discrete structure is real but non-observable: only emergent physical quantities are measurable. In informational models (Chiribella, D'Ariano & Perinotti 2011; Van Raamsdonk 2010; Swingle 2012), ontological correlations do not live in spacetime and are not observable, yet determine emergent physical properties. Within this framework, the PSE distinction between $S(\text{real})$, $D_S^{(f)}$ and $D_S^{(e)}$ (non-observable), and $\Phi: E_{reg}^{(k)}$ (observable) is fully consistent with contemporary literature on pregeometry and the emergence of physical structures.

1.2 Fundamental Structure

The fundamental layer of reality is described by the prespatial structure S , composed of elements lacking coordinates, metric, causality, or physical properties. The fundamental reorganization $\mathfrak{D}$ acts on S , generating:

- the **fundamental structural distance** $D_S^{(f)}$, discrete and non-physical;
- the **fundamental correlation** C_f , binary and selective, regulated by the coherence threshold Δ .

The fundamental domain is therefore a real, pregeometric, and non-observable structure.

1.3 Emergent Structure

From the aggregation of fundamental correlations emerges the **emergent structural distance** $D_S^{(e)}$, continuous and non-physical. On this distance is defined the **emergent correlation** C_e , a monotonic decreasing function of $D_S^{(e)}$.

These quantities define the **emergent domain** X : a continuous, non-observable domain devoid of physical quantities.

The combination of X , $D_S^{(e)}$, and C_e constitutes the **emergent structure** E . It is important to emphasize that **E is not an autonomous ontological domain**: it is a **structure defined on the emergent domain** X , obtained from the coherent organization of emergent quantities.

The emergent structure E is:

- continuous but non-physical,

- coherent but non-geometric,
- regular but non-observable.

It represents the **first continuous structural level of the theory**, but it does not yet possess the stability required to generate physical quantities.

Technical Note

*The distinction between the emergent domain X and the emergent structure E is consistent with contemporary models in which continuity is not a physical domain but a structure defined on a pregeometric substrate. In causal set theory (Bombelli et al. 1987; Dowker 2020), continuity emerges as an effective property of discrete relations but does not constitute an autonomous domain. In spin-network condensates (Gielen & Oriti 2016; Chirco & Oriti 2017), the pregeometric continuous structure is defined over a set of discrete states, analogous to how E is defined over X . In tensor network models (Swingle 2012; Qi 2013), distance and coherence emerge as functions of non-physical correlations, showing that a continuous structure can be defined on a non-physical domain. Within this framework, the emergent structure E of the PSE stands as a **continuous structure defined on the emergent domain**, not as a separate ontological domain.*

1.4 Regularization and Complexity Thresholds

The raw emergent structure is not yet compatible with physical projection. To acquire stability and compressive continuity, it must be regularized through:

- the **regularization operator R** ,
- the **complexity thresholds Θ_k** , which select sufficiently rich emergent configurations.

The result is the **regularized emergent structure**:

$$E_{\text{reg}}^{(k)} = \left(X, D_S^{(e)} \text{ reg}, C_e \text{ reg} \right)^{(k)}$$

Only this structure is compatible with the projection Φ .

Technical Note

The need to regularize an emergent structure before obtaining physical quantities is consistent with models in which physics arises from filtering, coarse-graining, or structural selection processes. In causal set theory (Sorkin 2003; Dowker 2020), only certain discrete configurations possess the complexity required to generate effective geometries. In tensor network models (Swingle 2012; Evenbly & Vidal 2011), physical continuity emerges only after renormalization operations that filter correlations and select stable structures. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physical dynamics is interpreted as an effective representation of regularized microscopic properties. Within this context, the regularization operator R and the complexity thresholds Θ_k of the PSE play an analogous role: they select emergent configurations endowed with the stability and coherence required for the projection Φ to formalize physical quantities.

1.5 Physical Projection

The monotonic map Φ formalizes part of the regularized emergent structure into the physical domain:

$$\Phi: E_{\text{reg}}^{(k)} \longrightarrow \mathcal{D}_{\text{fisico}}.$$

Φ does not create physics: it **represents** it. Physical quantities are compressed images of non-physical emergent quantities.

Within the physical domain appear:

- observables,
- geometric, dynamical, statistical, field-theoretic, or discrete structures,
- the quantum structure $(\mathcal{H}, A, \hbar)$,
- physical causality,
- experimental phenomenology.

Technical Note

The formalization of a pregeometric structure into physical quantities through a monotonic and non-invertible map is consistent with several contemporary emergentist approaches. In holographic models (Ryu & Takayanagi 2006; Van Raamsdonk 2010), geometric and dynamical quantities are reconstructed from quantum correlations via maps that compress non-local information, analogous to the role of Φ in the PSE. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physical quantities such as energy and dynamics are interpreted as effective representations of non-geometric microscopic properties, showing that physics can emerge from more primitive structures. In tensor network models (Swingle 2012; Qi 2013), the state space and geometric structures are compressed images of discrete correlations, demonstrating that physics can be obtained through monotonic maps that reduce information. Within this framework, the projection Φ of the PSE acts as a formal mechanism that translates the regularized emergent structure into observable physical quantities, without introducing new information and preserving the non-invertibility of the ontological hierarchy.

1.6 Synthesis and Ontological Chain

This chapter has introduced the ontological foundations of the PSE and defined the structures that compose the hierarchy of emergence:

- $S \rightarrow$ fundamental structure
- $D_S^{(f)}, C_f \rightarrow$ fundamental level
- $X, D_S^{(e)}, C_e \rightarrow$ emergent level
- $E, E_{\text{reg}}^{(k)} \rightarrow$ regularized emergent structure
- $\Phi \rightarrow$ physical projection
- **physical domain** $\rightarrow$ only observable level

These entities are organized into a **unidirectional and non-invertible ontological chain**:

$$S \xrightarrow{\mathcal{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} (X, D_S^{(e)}, C_e)_{\text{reg}} \xrightarrow{\Theta_k} (X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

In its simplified form:

$$S \xrightarrow{\mathcal{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

Each level contains **less information** than the previous one, and no physical quantity can reconstruct the prespatial structure from which it emerges. This chain constitutes the conceptual backbone of the PSE and will be developed further in the following chapters.

2. Structural Principle of Entanglement

2.1 Short Version

The fundamental layer of reality is a prespatial relational structure S , devoid of coordinates, metric, causality, and physical properties. The fundamental reorganization $\mathfrak{D}$ generates the fundamental structural distance $D_S^{(f)}$ and the fundamental correlation C_f which organize the fundamental domain.

The aggregation of fundamental correlations produces the emergent distance $D_S^{(e)}$ and the emergent correlation C_e which define the emergent domain X and the emergent structure E . When the emergent structure reaches sufficient complexity and is regularized, the monotonic projection Φ formalizes it into the physical domain, generating observable quantities.

Physical entanglement is the manifestation of the emergent correlation C_e compressed into the physical domain: it does not depend on physical distance, but on the regularized pregeometric structure.

Technical Note

The interpretation of physical structures as emergent from non-geometric correlations is consistent with several contemporary approaches to pregeometry. In holographic models (Ryu & Takayanagi 2006; Van Raamsdonk 2010), geometry is reconstructed from quantum correlations. In causal set theory (Bombelli et al. 1987; Sorkin 2003; Dowker 2020), physical structures emerge from discrete relations. In tensor network models (Swingle 2012; Qi 2013), continuous structures are images of informational networks. Within this framework, entanglement as the physical manifestation of structural proximity is fully consistent with the idea that non-local correlations arise from pregeometric structures.

2.2 Extended Version

The fundamental prespatial structure S is composed of elements lacking physical properties. The fundamental reorganization $\mathfrak{D}$ orders the configurations of S and generates the fundamental structural distance $D_S^{(f)}$, discrete and non-geometric. From $D_S^{(f)}$ emerges the fundamental correlation C_f , binary and selective, regulated by the coherence threshold Δ .

The collective aggregation of fundamental correlations produces the emergent distance $D_S^{(e)}$, continuous and non-observable, from which derives the emergent correlation C_e a monotonic decreasing function of $D_S^{(e)}$. These quantities define the emergent structure E which possesses continuity and coherence but is not yet compatible with a physical representation.

The emergent structure E must be regularized through the operator R and filtered by the complexity thresholds Θ_k which select sufficiently stable and rich emergent configurations. The result is the regularized emergent structure $E_{\text{reg}}^{(k)}$ the only one compatible with physical projection.

The monotonic projection Φ formalizes part of the regularized emergent structure into the physical domain, generating measurable quantities. Φ does not introduce new properties: it compresses the emergent structure into effective physical forms.

Physical entanglement is therefore the compressed image of the emergent correlation C_e : a physical property reflecting regularized pregeometric coherence.

Technical Note

The idea that quantum correlation arises from a pregeometric structure is consistent with foundational results in quantum information theory. Bell's theorem (1964) and the CHSH inequalities (Clauser, Horne, Shimony & Holt 1969) show that no local theory can reproduce quantum correlations. Tsirelson's bound (1980) establishes the maximum strength of quantum correlations compatible with the theory. In holographic models (Maldacena & Susskind 2013), entanglement is interpreted as a pregeometric structure determining connections between physical regions. Within this context, the emergent correlation C_e of the PSE stands as the structural basis of physical entanglement.

2.3 Ontological Hierarchy of Levels

The ontological hierarchy of the PSE is articulated into three distinct levels:

Fundamental Level

- prespatial structure S ,
- fundamental distance $D_S^{(f)}$,
- fundamental correlation C_f . It is discrete, real, and non-observable.

Emergent Level

- continuous domain X ,
- emergent distance $D_S^{(e)}$,
- emergent correlation C_e ,
- emergent structure E . It is continuous, coherent, and non-observable.

Physical Level

- arises exclusively through Φ ,
- contains geometric, dynamical, statistical, field-theoretic, discrete, and quantum structures,
- is the only observable level.

The hierarchy is strictly **unidirectional and compressive**: each level contains less information than the previous one. No physical quantity allows reconstruction of the emergent structure or the fundamental structure.

Technical Note

The ontological hierarchy of the PSE is consistent with emergentist models in which each level contains less information than the one preceding it. In causal set theory (Dowker 2020), physical structures contain less information than the original discrete structure. In holographic models (Ryu & Takayanagi 2006), geometry is a compression of quantum correlations. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physics is interpreted as a macroscopic phenomenon derived from microscopic non-geometric structures. Within this framework, the PSE hierarchy stands as a coherent, non-invertible, and non-retroactive emergent structure.

2.4 Fundamental Axioms

Axiom 1 — Structural Fundamentality

There exists a real prespatial structure S , devoid of coordinates, metric, causality, dynamics, and physical properties. The elements of S are defined exclusively by the structural relations that connect them. S constitutes the primary ontological level of the theory.

Axiom 2 — Fundamental Reorganization $\mathfrak{D}$

The prespatial structure S is subject to a fundamental reorganization $\mathfrak{D}$, which orders structural configurations through:

- a discrete structural index t , non-physical and non-metric;
- a structural coherence limit c , which constrains admissible variations;
- the $E + C$ principle, which stabilizes configurations by reducing tensions and contradictions.

The reorganization $\mathfrak{D}$ generates the fundamental structural distance $D_S^{(f)}$.

Axiom 3 — Fundamental Structural Distance and Correlation

On S is defined the fundamental structural distance $D_S^{(f)}$ discrete and pregeometric. From it derives the fundamental correlation C_f , binary and selective, regulated by the coherence threshold Δ . The aggregation of fundamental correlations forms the basis for the emergence of the emergent distance.

Axiom 4 — Emergent Structural Distance and Correlation

From the reorganization of fundamental correlations emerges the **emergent structural distance** $D_S^{(e)}$, continuous, regular, and compressive with respect to $D_S^{(f)}$. From $D_S^{(e)}$ derives the **emergent correlation** C_e , defined through a monotonic decreasing function:

$$C_e = F\left(D_S^{(e)}\right)$$

where F depends on the regularization class of the emergent structure. Both $D_S^{(e)}$ and C_e belong to the pregeometric domain and constitute the minimal emergent structures.

Axiom 5 — Regularization and Complexity Threshold

The raw emergent structure is not compatible with a physical representation. To acquire stability and compressive continuity, it must be regularized through:

- the regularization operator R ,
- the complexity thresholds θ_k which select sufficiently rich emergent configurations.

The result is the **regularized emergent structure**:

$$E_{\text{reg}}^{(k)} = \left(X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}} \right)^{(k)}$$

Axiom 6 — Physical Projection

Physical quantities emerge through a monotonic and compressive map:

$$\Phi: E_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{physical}}$$

Φ does not introduce new structures: it formalizes part of the regularized emergent structure into effective physical quantities. The physical domain is a compressed representation of the emergent level.

Axiom 7 — Emergent Physical Structures and Limits

Geometric, dynamical, statistical, field-theoretic, discrete, and quantum structures do not exist at the fundamental level. They appear in the physical domain as compressed images of $D_S^{(e)}$ and C_e . The structural limits of the fundamental level (coherence c , granularity h , minimal and total rigidity) appear in the physical domain as phenomenological constants and constraints.

Technical Note on the Axioms

The structural axioms of the PSE are consistent with pregeometric and informational approaches to quantum gravity. The axiom of structural fundamentality aligns with discrete models (Bombelli et al. 1987; Markopoulou 2000). The axiom of the structural index is coherent with timeless approaches (Barbour 1999) and with the thermal time hypothesis (Connes & Rovelli 1994). The axiom of fundamental correlation is in line with logical-relational models (Barwise & Seligman 1997) and informational frameworks (Chiribella, D'Ariano & Perinotti 2011). The axiom of emergent correlation is consistent with models where geometry arises from correlations (Van Raamsdonk 2010; Swingle 2012). The axioms on emergent physical structures and emergent limits align with models in which physical quantities derive from discrete structures (Hossenfelder 2018; Trugenberger 2017–2024).

3. Ontological Consequences

3.1 Fundamental and Emergent Structural Distance

Structural distance is an internal property of the prespatial structure S . It is not a geometric distance, does not presuppose space, time, causality, or metric structure, and does not correspond to any physical notion of separation. It is an ontological relation that measures **structural proximity** between elements or configurations of the structure. It appears in two distinct forms: a fundamental one and an emergent one.

Fundamental Structural Distance $D_S^{(f)}$

$D_S^{(f)}$ is the distance defined between the fundamental elements of the prespatial structure.

Characteristics:

- **Discrete:** it takes discrete, non-continuous values.
- **Pregeometric:** it is not defined within any spatial framework; it is an internal relation of S .
- **Non-physical:** it does not correspond to any observable distance.
- **Selective:** it determines which pairs of elements may contribute to the fundamental correlation C_f .
- **Relational:** it measures structural coherence between elements, not spatial positions.
- **Local:** the fundamental reorganization $\mathfrak{D}$ acts only on relations that are close in the sense of $D_S^{(f)}$.

In summary: $D_S^{(f)}$ is an ontological distance between fundamental elements, entirely devoid of geometric meaning.

Emergent Structural Distance $D_S^{(e)}$

$D_S^{(e)}$ is the distance defined between emergent configurations of the structure, i.e., coherent patterns composed of many fundamental elements.

Characteristics:

- **Aggregated:** it results from many fundamental relations.
- **Continuous:** it appears continuous as a collective effect of aggregation.
- **Coherent:** it measures structural proximity between stable emergent configurations.
- **Pregeometric:** it is not yet a physical distance; it belongs to the emergent but still non-observable level.
- **Derived:** it is constructed from $D_S^{(f)}$, the fundamental correlation C_f , and the selectivity threshold Δ .

In summary: $D_S^{(e)}$ is a distance between emergent patterns, not a distance in physical space.

Relation Between the Two Distances

The emergent distance is a transformation of the fundamental distance:

$$D_S^{(e)}(P, Q) = F(D_S^{(f)}(i, j))_{i \in P, j \in Q},$$

where F is an aggregation function consistent with the selectivity imposed by Δ . The threshold Δ determines which fundamental correlations are admissible and therefore which contributions enter the construction of $D_S^{(e)}$.

Associated Correlations

- **Fundamental correlation C_f** — binary, discrete, selective; defined on fundamental elements.
- **Emergent correlation C_e** — continuous, aggregated; defined on emergent configurations.

Both correlations are pregeometric and non-observable

Technical Note

The distinction between fundamental and emergent distance is consistent with models in which physical structures derive from discrete relations. In causal sets (Bombelli et al. 1987; Dowker 2020), fundamental distance is discrete, while continuous distance emerges from aggregation processes. In spin network models (Markopoulou 2000; Oriti 2016), fundamental distance is defined by discrete relations, while continuous distance is a collective effect. In informational models (Van Raamsdonk 2010; Maldacena & Susskind 2013), emergent distance is a function of correlations, analogous to the role of C_e in the PSE.

3.2 Threshold Δ and Correlation Selectivity

The threshold Δ is an ontological constraint internal to the prespatial structure S . It is not a numerical value, not a physical parameter, and cannot be tuned or assigned: Δ is a property of the structure, not a constant. Its role is to determine which fundamental relations possess sufficient structural coherence to be admissible.

Since the fundamental structural distance $D_S^{(f)}$ is discrete and lacks any geometric interpretation, Δ does not compare metric quantities and does not define thresholds such as “small” or “large.” Δ acts directly on the structural coherence of fundamental relations: a relation between two elements is

admissible only if the discrete configuration of $D_S^{(f)}$ satisfies the constraint imposed by Δ . In that case, the fundamental correlation is present ($C_f = 1$); otherwise, it is absent ($C_f = 0$.) Δ does not evaluate magnitudes or intensities: it selects exclusively discrete patterns of structural proximity.

Ontological Clarification

Δ is neither a function nor a numerical parameter: it is a **structural predicate** that determines whether a fundamental relation is admissible or not. Its action is logical-structural in nature (admissible / non-admissible), not quantitative. Δ defines the coherent domain of fundamental relations without introducing any physical measure or threshold, and determines the ability of the fundamental level to generate a regular emergent structure.

A fundamental relation is admissible only if its discrete configuration of $D_S^{(f)}$ belongs to the coherent domain defined by Δ . Only admissible fundamental correlations contribute to the construction of the emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e . The discreteness of the fundamental domain does not arise from any physical process, but from this structural constraint: Δ determines the very form of the fundamental level and its ability to generate a coherent emergent structure.

The selectivity imposed by Δ produces rigidity, saturation, and regularity in physical quantities, preventing uncontrolled divergences and ensuring that the physical domain is bounded and stable. This mechanism is analogous to structural filtering and informational coarse-graining processes, in which only significant correlations survive structural compression.

Δ ensures that the emergent structure is regular and non-pathological. The selectivity of correlations is a fundamental property of the ontological hierarchy, not a physical effect.

Technical Note

Correlation selectivity is consistent with models in which emergent structure derives from filtered or thresholded correlations. In holographic models (Van Raamsdonk 2010; Maldacena & Susskind 2013), only sufficiently strong correlations contribute to the reconstruction of emergent geometry. In tensor networks (Swingle 2012; Qi 2013), emergent distance is determined by the strength of correlations, which act as a structural threshold. In informational models (Chiribella & Scandolo 2015), distance is defined as a monotonic function of selected correlations. In coarse-graining emergence models (Bény 2013; Ojima 2004), selectivity is essential for obtaining regular, divergence-free structures. In this context, the PSE's Δ stands as an ontological threshold determining which fundamental correlations participate in constructing the emergent structure.

3.3 Fundamental and Emergent Correlation

The fundamental correlation C_f is binary, selective, and pregeometric. It is determined solely by the fundamental structural distance $D_S^{(f)}$ and regulated by the coherence threshold Δ , which specifies which fundamental relations are admissible. C_f has no autonomous dynamics: every variation of C_f arises exclusively from local reorganizations of the fundamental structural distance. It is not a physical quantity, not observable, and does not constitute a hidden variable.

The emergent correlation C_e is defined by:

$$C_e(i, j) = F \left(D_S^{(e)}(i, j) \right),$$

where F is a monotonic function that translates the emergent distance into a degree of structural correlation.

Terminological Note

The function F used to define the emergent correlation C_e is a monotonic function operating on the emergent domain. It must not be confused with the **emergent operators** F_λ which instead act on the fundamental domain and represent the condition of regularizability along the approximation scale λ .

Characteristics of C_e :

- **continuous**, because it aggregates many fundamental relations;
- **regular**, because it inherits the coherence of $D_S^{(e)}$;
- **pregeometric and non-observable**, because it belongs to the emergent non-physical domain;
- **structural**, because it does not yet represent any physical quantity.

C_e does not represent any attenuated or preliminary form of physical correlation: it is a purely structural correlation, devoid of physical meaning until the projection Φ . Only Φ makes part of the emergent correlation physically describable, without altering its ontological nature.

Ontological role:

- it links the fundamental distance $D_S^{(f)}$ to the physical domain;
- it determines physical quantities through the monotonic map Φ ;
- it provides the structural basis for physical entanglement, which emerges as a compressed image of non-observable emergent correlations.

Technical Note

The interpretation of C_e as a monotonic function of emergent distance is consistent with models in which physical structures derive from correlations. In holographic models (Van Raamsdonk 2010; Maldacena & Susskind 2013), physical structures are reconstructed from monotonic correlations. In tensor networks (Swingle 2012; Qi 2013), correlation is the basis of emergent distance. In informational models (Chiribella & Scandolo 2015), distance is defined as a monotonic function of correlations. In this context, the PSE's C_e stands as a conceptual bridge between prespatial structure and the physical domain.

3.4 Emergent Operators F_λ and G_λ

The emergence of continuous structural quantities in the emergent domain implies the existence of two families of regularization operators, F_λ and G_λ , which act on the discrete fundamental quantities $D_S^{(f)}$ and C_f . These operators do not represent any physical or geometric dynamics: they are purely structural transformations expressing the condition of regularizability of the fundamental quantities along the approximation scale λ .

The operator F_λ acts on the fundamental structural distance:

$$F_\lambda: D_S^{(f)} \rightarrow D_\lambda,$$

where D_λ is a regularized distance defined on the approximated domain S_λ . Similarly, the operator G_λ acts on the fundamental correlation:

$$G_\lambda: C_f \rightarrow C_\lambda,$$

where C_λ is a regularized correlation defined on the same domain.

The families $\{F_\lambda\}_{\lambda \in \Lambda}$ and $\{G_\lambda\}_{\lambda \in \Lambda}$ are monotonic, compressive, and stable: for $\lambda_2 > \lambda_1$, the quantities $F_{\lambda_2}(D_S^{(f)})$ and $G_{\lambda_2}(C_f)$ are more regular than the corresponding approximations at smaller scales. Compressivity ensures that each transformation reduces the information contained in the fundamental quantities, while stability guarantees that local variations compatible with the structural bound

$$|D_S^{(f)}(t+1) - D_S^{(f)}(t)| \leq c$$

produce controlled variations in the regularized quantities D_λ and C_λ .

The limits

$$D_S^{(e)} = \lim_{\lambda \rightarrow \infty} F_\lambda(D_S^{(f)}), \quad C_e = \lim_{\lambda \rightarrow \infty} G_\lambda(C_f)$$

respectively define the emergent structural distance and the emergent correlation of the domain X . These quantities are continuous, pregeometric, non-observable, and saturated: they do not represent any physical magnitude, but constitute the mathematical basis for the future projection Φ that will generate the physical domain.

The emerging operators F_λ and G_λ therefore express the ontological condition through which the fundamental structure becomes regularizable and continuous in the limit, preparing the emergence of the structural quantities $D_S^{(e)}$ and C_e that define the non-observable emergent domain X

3.5 No Physical Dynamics in the Fundamental Levels

In the fundamental and non-observable emergent levels, no form of physical dynamics exists. In these domains:

- no physical equations of motion exist;
- no physical causality exists;
- no physical propagation exists;
- no metric exists;
- no energy exists;
- no mass exists;
- no gravity exists.

The transformations occurring in the fundamental domain are purely structural: they concern only the discrete relations of the prespatial structure S and their internal coherence. No physical quantity, no geometry, and no observable dynamics exist at this level. The discreteness of the fundamental domain and the selectivity imposed by Δ determine the form of the emergent level but do not introduce any physics.

Physics arises only when the regularized emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ reaches the required complexity and is projected into the physical domain through the monotonic map Φ , which formalizes the emergence of observables.

Technical Note

The absence of fundamental physical quantities is consistent with timeless approaches, non-commutative geometry, informational models, and spin-network quantum gravity. Within this framework, the PSE places physics as an emergent phenomenon, not a primitive one.

4. Architecture of the Domains and of the Emergent Structure

The theory distinguishes two ontological domains (fundamental and emergent), a continuous emergent structure defined on the emergent domain, and an observable physical domain that does not possess ontological nature. Each level emerges from the previous one through non-invertible and non-retroactive transformations. This articulation is essential to avoid overlaps between levels and to clarify which quantities belong to the prespatial structure and which to the physical representation.

4.1 Fundamental Non-Observable Domain

The fundamental domain is the first ontological level of the theory. It is composed of the prespatial structure S , the fundamental structural distance $D_S^{(f)}$, and the fundamental correlation C_f . Its main features are:

- absence of geometry;
- absence of physical causality;
- absence of continuity;
- non-observability;
- description through a discrete, non-physical structural index;
- presence of the structural coherence limit c .

In this domain, only **discrete structural relations** exist between elements lacking coordinates or physical properties. The fundamental reorganization $\mathcal{D}$ acts locally on the relational configurations of S , generating the fundamental structural distance $D_S^{(f)}$. From $D_S^{(f)}$, through the coherence threshold Δ , the fundamental correlation C_f emerges, binary and selective.

No physical quantity belongs to this level: the fundamental domain is real but completely prespatial and non-observable. The selectivity imposed by Δ introduces a **structural rigidity** that is inherited by the emergent levels, preventing instabilities in the construction of $D_S^{(e)}$ and C_e .

Technical Note

The distinction of a fundamental domain devoid of physical quantities is consistent with several contemporary approaches to pregeometry. In causal set theory (Bombelli et al. 1987; Sorkin 2003; Dowker 2020), the microscopic level consists of discrete elements without coordinates, and physical structures emerge only at an effective level. In non-commutative geometry (Connes 1994; Connes & Marcolli 2008), the fundamental structure is algebraic and does not possess physical quantities: such properties emerge from spectral operators. In dynamical spin-network models (Markopoulou 2000; Oriti 2016), the fundamental state is discrete, non-metric, and non-observable, and the dynamics acts on elementary relations. Within this framework, the fundamental domain of the PSE is positioned as a discrete prespatial structure, devoid of physical quantities and governed by a local fundamental reorganization acting on $D_S^{(f)}$.

4.2 Emergent Non-Observable Domain X

From the fundamental domain there emerges a continuous and regular level, constituted by the **emergent domain X** . X is a pregeometric, non-observable domain, devoid of physical quantities, and represents the **ontological support** on which the continuous structural quantities of the emergent level can be defined.

The emergent domain introduces continuity into the hierarchy, but it is not sufficient on its own to fully describe the emergent level: for this reason, it is necessary to consider the structural quantities defined on it, which will be introduced in the next paragraph as the **emergent structure E** .

Technical Note

The existence of a continuous yet non-observable emergent domain is consistent with models in which continuity arises as a collective effect of discrete structures. In causal set theory (Bombelli et al. 1987; Dowker 2020), continuity emerges from coarse-graining processes but is not yet physical. In spin-network condensate models (Gielen & Oriti 2016; Chirco & Oriti 2017), a continuous pregeometric level precedes emergent physics. In informational models (Van Raamsdonk 2010; Maldacena & Susskind 2013), continuity derives from the structure of correlations, while physical quantities emerge only after an effective projection. Within this framework, the emergent domain of the PSE is positioned as a continuous, coherent, and non-physical pregeometric level.

4.3 Emergent Non-Observable Structure E

The emergent structure E is the first continuous structure of the theory. It does not constitute an autonomous ontological domain: it is a structure defined on the emergent domain X , obtained from the coherent aggregation of the fundamental quantities. E combines three elements:

- **The emergent domain X** : a continuous, non-observable domain, devoid of geometry and physical quantities.
- **The emergent structural distance $D_S^{(e)}$** : a continuous function derived from the aggregation of many fundamental relations. It is not a geometric distance, but a coherent and non-physical structural distance.
- **The emergent correlation C_e** : a monotonically decreasing function of $D_S^{(e)}$, continuous and non-physical, representing the emergent coherence of the prespatial level.

These three elements form the complete emergent structure:

- continuous but non-physical,
- coherent but non-geometric,
- regular but non-observable,
- self-sufficient and devoid of physical dynamics.

The emergent structure E is therefore the first continuous and coherent organization of prespatial reality. It conceptually mediates between the fundamental structure and the physical representation, but it is not yet usable to define physical quantities.

Regularization of the Emergent Structure

The emergent structure E does not possess the stability and complexity required to generate physical quantities. For this reason, it must be regularized through:

- the regularization operator R ,
- the complexity thresholds Θ_k .

Regularization yields:

$$E_{\text{reg}}^{(k)} = \left(X, D_S^{(e)} \text{ reg}, C_e \text{ reg} \right)^{(k)}$$

Only when the emergent structure assumes this regularized and filtered form does it become possible to represent it through the monotonic map Φ , which formalizes the effective physical quantities.

Properties of the Emergent Structure E

- $D_S^{(e)}$ is an aggregated function of many fundamental relations.
- C_e is a monotonically decreasing function of $D_S^{(e)}$.
- no quantity is observable;
- no physical structure exists;
- the emergent correlation is coherent but non-geometric.

The emergent structure E constitutes the conceptual bridge between the prespatial structure and the physical representation obtained through Φ .

Technical Note

The emergent structure E , as a combination of a continuous domain and non-physical structural quantities, is consistent with several contemporary emergentist models. In causal set theory (Bombelli et al. 1987; Dowker 2020), continuity is not a physical domain but an effective structure derived from discrete relations. In spin-network condensate models (Gielen & Oriti 2016; Chirco & Oriti 2017), a continuous pregeometric structure precedes observable physics. In tensor-network models (Swingle 2012; Qi 2013), distance and coherence emerge as monotonic functions of non-physical correlations. Within this framework, the emergent structure of the PSE is positioned as a continuous, coherent, and non-observable level that prepares the physical representation without possessing any dynamics or physical quantities.

4.4 Observable Physical Domain

The observable physical domain arises from the monotonic projection Φ applied to the regularized emergent structure:

$$E_{\text{reg}}^{(k)} = \left(X, D_S^{(e)} \text{ reg}, C_e \text{ reg} \right)^{(k)}$$

When $E_{\text{reg}}^{(k)}$ exceeds a complexity threshold sufficient to produce stable, coherent, and regular configurations, it becomes possible to represent it through Φ . Φ does not generate physical quantities: **it formalizes them**, translating the pregeometric emergent structure into effective physical magnitudes:

$$\Phi: E_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{phys}}.$$

The stability and regularity of the emergent physical quantities reflect the structural rigidity induced by Δ , which restricts the admissible fundamental correlations and prevents divergences in the emergent structure.

Within the physical domain appear:

- observable physical quantities;
- geometric, dynamical, statistical, field-theoretic, or discrete structures;
- the quantum structure $(\mathcal{H}, A, \hbar)$ as a compressed image of the emergent correlational constraints;
- physical causality as an effective representation of emergent coherence;
- experimental phenomenology.

Properties of the Physical Domain

- it is the only observable domain;
- it possesses physical structures;
- physical causality is an effective representation of emergent correlation formalized by Φ ;
- the relativistic limit is the compressed image of the structural coherence limit c ;
- physics is an effective representation of $D_S^{(e)}$ and C_e not a reconstruction of the emergent structure.

Technical Note

Interpreting the physical domain as a compressed representation of a pregeometric structure is consistent with several emergentist approaches. In holographic models (Ryu & Takayanagi 2006; Van Raamsdonk 2010), physical structures are reconstructed from correlations through monotonic, non-invertible maps. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physical structures are effective representations of microscopic non-geometric properties. In tensor-network models (Swingle 2012; Qi 2013), physical structures emerge as continuous limits of discrete networks. Within this framework, the physical domain of the PSE is positioned as an observable level made accessible by emergent complexity and formalized by Φ .

4.5 Relations Between Domains

The hierarchy is:

$$S \xrightarrow{\mathcal{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} E_{\text{reg}} \xrightarrow{\Theta_k} E_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{phys}}$$

Properties of the Hierarchy

- **unidirectional:** no level retroacts on the previous one;
- **non-invertible:** no physical quantity can reconstruct the prespatial structure;
- **non-local:** emergent correlation is not constrained by any physical structure;
- **non-physical in the initial levels:** only the domain generated by Φ is observable;
- **coherent:** each level inherits regularity from the previous one.

This structure prevents physicalist interpretations of the fundamental level and clarifies that physics is an **emergent representation**, not a set of primitive axioms.

Technical Note

The unidirectional and non-invertible hierarchy between levels is consistent with models in which physical structures cannot reconstruct the microscopic structure. In causal set theory (Sorkin 2003; Dowker 2020), emergent structures do not allow one to recover the original discrete configuration. In holographic models (Ryu & Takayanagi 2006; Van Raamsdonk 2010), the maps that reconstruct physical structures from correlations are monotonic and non-invertible. In emergent gravity models (Jacobson 1995; Padmanabhan 2010), physical structures do not retroact on the microscopic level. In informational models (Chiribella, D'Ariano & Perinotti 2011), emergent correlation does not live in any physical domain. Within this framework, the PSE hierarchy is positioned as a coherent and non-retroactive ontological structure.

Technical Note on the Quantum Structure

The quantum structure $(\mathcal{H}, A, \hbar)$ does not constitute an additional ontological domain: it is an effective representation obtained through the compression map Φ_{quant} , which translates the correlational constraints of the emergent domain into states, operators, and minimal thresholds. Quantization is not a fundamental property of reality, but an effect of the saturation of emergent correlation and of the compressive nature of the projection.

This interpretation is consistent with several contemporary approaches to emergent quantum mechanics. In informational models (Chiribella, D'Ariano & Perinotti 2011), quantum states are effective representations of more primitive correlational structures. In holographic models (Maldacena 1997; Van Raamsdonk 2010), quantum structure emerges from monotonic maps that compress non-local correlations. In tensor-network models (Swingle 2012; Qi 2013), the state space is an effective representation of discrete correlational constraints. Within this framework, the quantum structure of the PSE is positioned as a **compressed image of emergent correlation**, not as a separate ontological level.

5. Fundamental Reorganization $\mathfrak{D}$

The fundamental reorganization $\mathfrak{D}$ is the structural process that rewrites the prespatial structure S along the discrete structural index t . It acts exclusively on the fundamental structure S , reorganizing the relations among elements that possess no physical properties. “The ‘action’ of $\mathfrak{D}$ does not imply any dynamics, evolution, or temporal process: $\mathfrak{D}$ does not modify S over time, but defines which configurations of the structure are admissible, coherent, and compatible with the ontological constraints. The reorganization is therefore a structural constraint, not a physical change.

The reorganization is local within the structure generated by $D_S^{(f)}$: admissible variations between adjacent configurations are limited by the structural coherence parameter c , which introduces no notion of physical velocity or propagation. $\mathfrak{D}$ follows the structural principle $E + C$, which minimizes contradictions and reduces tensions among nearby relations.

$\mathfrak{D}$ is not a physical dynamics, not geometric, and not defined in any physical domain: it is a pregeometric structural process that precedes every form of physics and constitutes the primary ontological level from which all subsequent structures emerge.

5.1 Structural Index t

The structural index t is a discrete parameter that orders the fundamental configurations of the prespatial structure:

$$S_t \text{ with } t \in \mathbb{N}$$

It is not a physical time, it has no metric, it is not continuous, and it does not define causality. It serves exclusively to label the configurations of the prespatial structure. The index t does not represent any form of evolution: it is a simple ordering, devoid of temporal or dynamical meaning. Referring to “configurations along t ” only indicates the adjacency relation between admissible configurations, not a process in time. The fundamental reorganization $\mathfrak{D}$ is defined as a local property with respect to $D_S^{(f)}$: every variation of the structure involves only structural relations that are close in the sense of $D_S^{(f)}$. This reorganization is a property of the structure, not a process in time.

Technical Note

The interpretation of the structural index t as an internal property of the pregeometric structure is consistent with several contemporary approaches to pregeometry and rewriting dynamics. In such models, the fundamental transformation is described as a rewriting of states, devoid of any physical temporal variable. In computational models, the transformation is represented as a discrete sequence of local updates; in the thermal time hypothesis, observable time is interpreted as an effective parameter derived from internal transformations of the state. In this context, the PSE’s structural index introduces no ontological dynamics: it orders the configurations of the prespatial structure without defining succession, duration, or causality. Physical time arises only after the projection Φ , as a compressed image of the emergent structure.

5.2 Structural Coherence Limit c

The fundamental reorganization $\mathfrak{D}$ is subject to a structural coherence limit c , which regulates the amplitude of admissible modifications of the fundamental structural distance $D_S^{(f)}$. The limit c arises from the discrete nature of the prespatial structure S : since S is composed of elementary units and rewriting rules with finite support, every transformation between adjacent configurations can involve only a finite number of elementary units. It follows that there exists a maximum extent for the admissible reorganizations in a single ontological step.

Since S possesses no physical time, this reorganization is not a temporal evolution: it is a discrete structural property. The parameter c is therefore the fundamental constant that constrains the maximum amplitude of transformations between configurations of the prespatial structure. Its universality in the physical domain derives from the fact that c is an intrinsic property of the discrete prespatial structure, not of the physical domain.

Characteristics of c :

- it is not a physical velocity;
- it is not defined in any physical domain;
- it does not imply causality;
- it is a constraint on the reorganization of the fundamental structure.

The relativistic limit observed in the physical domain is the image of c through the projection Φ .

Technical Note

The interpretation of the fundamental limit c as a structural constraint is consistent with several contemporary approaches to discrete pregeometry. In discrete models, state transformations are limited by the granularity of the substrate; in causal sets, causality and the limiting speed emerge from discreteness, not from fundamental metric properties. Lieb–Robinson bounds show that there exists a maximum speed of correlational influence in discrete systems, independently of the presence of a

physical metric. In dynamic spin-network models, analogous limits emerge from the combination of discreteness and locality.

In this context, the PSE's structural limit c does not represent a physical velocity nor a fundamental dynamics: it is an internal constraint on the discrete reorganization of the prespatial structure, limiting the amplitude of admissible variations between adjacent configurations in the structural index. Its image in the physical domain, through the projection Φ , coincides with the observable relativistic constant.

5.3 Principle $E + C$

The fundamental reorganization $\mathfrak{D}$ is governed by a structural minimization principle:

$$E + C = \text{reduction of contradictions} + \text{reduction of tensions.}$$

- E represents the internal coherence of the structure: the reorganization tends to eliminate incompatible configurations.
- C represents local compatibility: the reorganization tends to reduce tensions among nearby relations.

This principle is not physical: it does not minimize energy, does not define equations of motion, and does not imply temporal continuity. It is a structural criterion that guides the reorganization of $D_S^{(f)}$.

Technical Note

The $E + C$ principle finds analogies in models of self-organization, complex networks, and logical-relational frameworks. In such contexts, stability emerges from the reduction of internal incompatibilities and from local compatibility, without relying on fundamental physical quantities. The PSE's $E + C$ principle stands as a general structural rule governing the reorganization of $D_S^{(f)}$.

5.4 Reorganization of $D_S^{(f)}$

The fundamental reorganization $\mathfrak{D}$ produces local transformations of the structure, rewriting the fundamental structural distance between adjacent configurations of the discrete structural index. Each transformation:

- modifies the fundamental correlation C_f ;
- alters local coherence;
- contributes to the emergence of $D_S^{(e)}$;
- does not generate any direct physical phenomena.

The fundamental reorganization is discrete, non-continuous, and non-physical. It is the process from which emergent regularity arises.

Technical Note

The local reorganization of the prespatial structure presents analogies with models of emergent geometry from discrete structures, coarse-graining processes, and quantum network transformation models. In such frameworks, continuity and coherence emerge from discrete local transformations. In this context, the PSE's fundamental reorganization is not an ontological dynamics: it is a local structural property of the fundamental distance $D_S^{(f)}$, ensuring the coherence of configurations

ordered by the discrete structural index. This property produces the continuity of the emergent distance $D_S^{(e)}$ without introducing succession, causality, or evolution in the fundamental domain.

5.5 Emergence of $D_S^{(e)}$

The emergent structural distance $D_S^{(e)}$ arises from the collective regularization of the fundamental reorganizations.

Characteristics:

- it is continuous;
- it is coherent;
- it is non-observable;
- it is non-physical;
- it is the basis of the emergent correlation C_e .

The fundamental reorganization does not act directly on $D_S^{(e)}$: it is an emergent effect of the reorganization of $D_S^{(f)}$.

Technical Note

The idea that a continuous distance may emerge from a discrete structure is supported by causal sets, emergent-geometry models based on correlations, and spin-network condensate approaches. In this context, the PSE's $D_S^{(e)}$ stands as a collective regularization effect, non-observable and non-physical.

5.6 Emergence of the Emergent Correlation C_e

The emergent correlation is a monotonically decreasing function:

$$C_e(i, j) = F\left(D_S^{(e)}(i, j)\right).$$

The fundamental reorganization $\mathfrak{D}$ does not act on C_e : variations of C_e derive exclusively from variations of $D_S^{(e)}$.

Technical Note

Models of emergent geometry based on correlations, holographic approaches, and relational informational models show that physical structures can be derived from monotonic functions of correlation. In this context, the PSE's C_e stands as a pregeometric structure derived from $D_S^{(e)}$.

6. Emergency Maps and Physical Representation

This chapter describes how the emergent structure generated by the fundamental domain is made compatible with a physical representation through two key steps: the regularization performed by the Structural Regularity Operator R and the application of the complexity thresholds Θ_k . Only after these filtrations does the emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ reach the stability, continuity, and minimal complexity required to be compressed into physical quantities through the emergency maps. The chapter then introduces the six fundamental classes of maps — discrete, geometric, dynamic, statistical, field, and quantum — showing how each provides an effective mode of translating the pregeometric structure into the physical domain. In this framework, physics is not an ontological level

but the set of compressed representations of the emergent correlation, from which geometry, dynamics, fields, constants, and all observable quantities derive.

6.1 Regularization of the Emergent Domain X

Within the conceptual framework of the PSE, the emergent domain is defined as:

$$X = \Psi(S),$$

where S is the fundamental structure. The emergence map

$$\Psi: S \rightarrow X$$

is the composition of the transformations that generate the emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e :

$$\Psi = G_\lambda \circ F_\lambda \circ \Delta \circ \mathcal{D},$$

namely:

$$S \xrightarrow{\mathcal{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e.$$

The result of these transformations is the **emergent structure**:

$$E = (X, D_S^{(e)}, C_e)$$

Although $D_S^{(e)}$ and C_e are already continuous, regular, and saturated, the emergent structure E is not yet compatible with the physical compression performed by the projection Φ . To ensure compressive continuity, global stability, and structural coherence, we introduce the **Structural Regularity Operator** R :

$$R: E \rightarrow E_{\text{reg}} = (X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}})$$

The operator R does not modify the domain X : regularization acts exclusively on the emergent quantities $D_S^{(e)}$ and C_e .

6.2 Complexity Thresholds of the Emergent Structure

Structural regularization yields a continuous and stable emergent structure,

$$E_{\text{reg}} = (X, D_{S \text{ reg}}^{(e)}, C_{e \text{ reg}}),$$

but it does not yet guarantee that this structure possesses the **minimum complexity** required for a physical map Φ to be well-defined. For this reason, we introduce the **complexity thresholds**:

$$\Theta_k: E_{\text{reg}} \rightarrow E_{\text{reg}}^{(k)}$$

which act as **structural filtering operators**. Each threshold Θ_k imposes a set of minimal requirements on the emergent structure, including:

- **topological complexity** (absence of local degeneracies),
- **correlative complexity** (minimum saturation of C_e),
- **metric complexity** (non-collapse of $D_S^{(e)}$),

- **informational complexity** (ability to support a monotone, non-invertible compression).

The thresholds do not modify the domain X , nor do they introduce new information: they are **monotone, stable, and non-amplifying operators** that select the portion of the emergent structure compatible with a physical representation.

Applying Θ_k yields an emergent structure satisfying:

$$E_{\text{reg}}^{(k)} = \Theta_k(E_{\text{reg}}),$$

which constitutes the **minimal ontological level** at which a physical map

$$\Phi: E_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{phys}}$$

is well-defined, continuous, compressive, and informationally rigid.

The complexity thresholds therefore represent the **final bridge** between the pregeometric emergent structure and the possibility of a coherent physical representation. Without Θ_k , the projection Φ would not be universally definable.

6.3 Emergence Maps

In the PSE, maps are not ontological entities: they are **structural relations** that connect different levels of the hierarchy without introducing any new fundamental information. The operators F_λ , G_λ , R , and the complexity thresholds Θ_k act within the non-observable domain, selecting, stabilizing, and regularizing the emergent structure. The projection map Φ represents the final step of the chain: it is a **non-ontological**, monotonic, compressive map that translates the regularized emergent domain into observable physical quantities.

In the PSE, physical quantities emerge through a monotonic map:

$$\Phi: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{phys}}.$$

Φ is not unique: different classes of maps exist, each generating a distinct physical domain. Geometry is only one of the possible effective representations of the emergent structure.

Φ is neither an ontological object nor a physical process: it is a **non-ontological representational relation** that formalizes the emergent structure without modifying it, translating it into physical quantities through a monotonic and non-invertible compression.

Φ has four universal properties:

- **Monotonicity**

$$D_S^{(e)}(i, j) < D_S^{(e)}(k, l) \Rightarrow \Phi(i, j) < \Phi(k, l)$$

- **Non-invertibility**

$$\Phi^{-1} \text{ does not exist}$$

- **Informational compressivity**

$$I \left(\Phi \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \right) < I \left(D_S^{(e)}, C_e \right)$$

- **Informational rigidity** No physical quantity can indefinitely amplify the emergent information.

6.4 Classes of Emergence Maps

In the PSE there exist exactly six fundamental classes of emergence maps. Each of them projects the pregeometric structure $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$ into a different emergent physical domain. All other maps appearing in the framework (for example those generating energy, symmetries, or other physical quantities) do not constitute new classes, but are **derived maps**, obtained as specializations or compositions of the six fundamental classes.

The six fundamental classes are:

$$\{\Phi_{\text{disc}}, \Phi_{\text{geom}}, \Phi_{\text{dyn}}, \Phi_{\text{stat}}, \Phi_{\text{field}}, \Phi_{\text{quant}}\}$$

They represent all the ways in which the emergent structure can generate effective physical domains, **with the sole exception of Φ_{disc}** . The discrete map is a special class: it does not generate a physical domain, but a **pre-physical representation** (graphs, networks, discrete structures) of the emergent correlation. Φ_{disc} is therefore a fundamental map, but not a physical one: it provides a structural reading of the pregeometry, from which the other maps can derive effective physical representations.

(1) Discrete Maps

$$\Phi_{\text{disc}}: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)} \rightarrow (G, \Gamma, \mathcal{N})$$

The discrete map Φ_{disc} is the only map that does not lead to the physical domain: it produces a discrete structural representation of the emergent level, useful for informational and discrete models, but devoid of physical meaning.

Technical Note: Causal sets, spin networks, informational models, and tensor networks show that graphs and networks can emerge from more primitive correlations. The discrete map of the PSE fits within this line: G, Γ , and $\mathcal{N}$ are compressed images of $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$.

(2) Geometric Maps

$$\Phi_{\text{geom}}: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)} \rightarrow (d, g_{\mu\nu}, \mathcal{C})$$

Technical Note: Geometry as an effective representation of a pregeometric structure is consistent with holographic approaches, tensor networks, causal sets, and emergent gravity. In these frameworks, distance, metric, and causality are not fundamental but emerge from more primitive correlations. The geometric map of the PSE fits within this line: $d, g_{\mu\nu}$, and $\mathcal{C}$ are compressed images of $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$, while the fundamental structure remains devoid of coordinates, metric, and physical causality.

(3) Dynamical Maps

$$\Phi_{\text{dyn}}: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)} \rightarrow (\hat{H}, \hat{U}(T), \rho)$$

Technical Note: Operational approaches, axiomatic reconstructions, relational formulations, and timeless models show that dynamics can emerge from deeper informational structures. The dynamical map of the PSE fits within this line: $\hat{H}, \hat{U}(T)$, and ρ are not fundamental entities, but effective representations of $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$.

(4) Statistical Maps

$$\Phi_{\text{stat}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow (p(x), S, \langle O \rangle)$$

Technical Note: Informational models, entropic gravity, black-hole phenomenology, and holographic approaches show that entropy and distributions can emerge from non-geometric structures. The statistical map of the PSE fits within this line: $p(x)$, S , and $\langle O \rangle$ are compressed images of $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$.

(5) Field Maps

$$\Phi_{\text{fields}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow (\phi(x), A_\mu(x), F_{\mu\nu})$$

Technical Note: Group Field Theory, spin networks, spin foams, informational models, and non-commutative geometry show that fields can emerge from discrete or algebraic structures. The field map of the PSE fits within this line: ϕ , A_μ and $F_{\mu\nu}$ are effective representations of $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$.

(6) Quantum maps

$$\Phi_{\text{quant}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow (\mathcal{H}, A, \hbar)$$

Technical note: Informational approaches, operational reconstructions, algebraic formulations and emergent-quantum models show that quantum structure can be interpreted as an effective representation of deeper correlational patterns. In these frameworks, the state space $\mathcal{H}$, the action A and the minimal threshold $\hbar$ are not fundamental entities, but compressed images of $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$. The quantum map of the PSE fits within this line: $\mathcal{H}$, A and $\hbar$ are effective representations of the emergent structure, while the fundamental domain remains devoid of operators, non-commutativity and primitive quantization.

6.5 Order of Emergence of the Map Classes

Physical reality does not emerge all at once, but through a continuous progression of complexity that begins at the fundamental structure and unfolds into natural thresholds of emergence. The emergent structure $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$ represents the first level at which identities, domains, and coherent correlations appear. As complexity increases, these configurations make possible different forms of representation: first **discreteness** (Φ_{disc}), which captures the ontological separation between regions; then **geometry** (Φ_{geom}), which describes continuity and the emergent metric; next **dynamics** (Φ_{dyn}), which formalizes the coherent evolution of configurations; subsequently **statistics** (Φ_{stat}), which represents collective phenomena; and finally **classical fields** (Φ_{field}), which describe interactions over continuous domains. Only when complexity reaches its maximal threshold does **quantum compression** (Φ_{quant}) become possible, introducing states, operators, a minimal action scale, and stable dynamical modes. Quantum structure is not a separate ontological level, but the final effective representation of the same emergent structure that generated the classical maps. Once they appear, quantum phenomena build the entire physical reality—from atoms to cosmic structures—projecting themselves into the spacetime generated by Φ_{geom} , yet without being ontologically contained within

it. Reality is therefore unified and progressive, and the maps are not distinct worlds but **representation thresholds** along the growth of emergent complexity.

6.6 Relationship Between the Map Classes

The emergence order of the map classes does not define an ontological hierarchy nor a functional composition among the maps, but reflects the complexity thresholds of the emergent structure

$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$. Each map Φ_i becomes definable only when the emergent structure reaches the minimal threshold Θ_i required by the domain it represents. The maps do not operate on one another: they all act directly on the same emergent triple and provide compressed representations of it in different physical domains.

The mathematical relationship among the maps is therefore a threshold relation,

$$\Theta_{\text{disc}} < \Theta_{\text{geom}} < \Theta_{\text{dyn}} < \Theta_{\text{stat}} < \Theta_{\text{campo}} < \Theta_{\text{quant}}$$

which expresses the natural progression of complexity required for each map to be well-defined. The maps are compatible sections of the total projection Φ : they do not derive from one another, they do not compose, and they do not form ontological levels, but represent different modes of monotonic compression of the same emergent structure.

6.7 The Role of Quantum Theory in the Hierarchy of Maps

The PSE introduces a decisive conceptual shift in the understanding of the levels of reality. Quantum theory does not belong to the fundamental domain, nor does it represent the original structure of the real: it is the most complex section of the map Φ , since its emergence requires all preceding projections. The quantum representation presupposes a geometric structure, an effective dynamics, a coherent statistical framework, a field-theoretic formalization, and a discrete basis; only after these steps can the emergent correlation be compressed into the form of states, operators, and minimal thresholds.

In this sense, quantum theory occupies a high level in the hierarchy of the Φ maps not because it is “deeper,” but because it is the most compressed and information-dense representation of the emergent correlation. Physics interprets the microscopic domain as an access to the fundamental, but in the PSE it is simply the region of the physical domain in which the compression performed by Φ_{quant} is maximal.

Quantum non-locality is not an intrinsic physical property, nor a propagative phenomenon in space: it is the compressed manifestation of the emergent correlation C_e , filtered through the mathematical structure of the quantum map. The quantum effects observable in the microscopic domain — entanglement, non-local correlations, indeterminacy — do not arise from fundamental dynamics, but from the physical representation of a pregeometric correlation.

Quantum theory is therefore a high-complexity emergent level within the hierarchy of the Φ maps, not the foundation of reality.

6.8 Emergent Physical Quantities

A physical quantity is defined as:

$$Q_{\text{physical}} = \Phi \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

No physical quantity is fundamental. Every observable magnitude is an informational compression of the emergent structure.

6.9 Special Cases

(a) Metric

$$g_{\mu\nu} = \Phi_{\text{geom}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

(b) Physical Time

$$T = \Phi_{\text{geom}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \quad \text{or} \quad T = \Phi_{\text{dyn}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

(c) Physical Dynamics

$$\text{dynamics} = \Phi_{\text{dyn}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

(d) Physical Fields

$$\phi(x) = \Phi_{\text{fields}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

(e) Statistical Distributions

$$P = \Phi_{\text{stat}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

(f) Symmetries

$$\mathcal{G} = \Phi_{\text{sym}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

with

$$\Phi_{\text{sym}} \subset \{ \Phi_{\text{geom}}, \Phi_{\text{dyn}}, \Phi_{\text{stat}} \}$$

(g) Energetic Structures

$$E = \Phi_{\text{ener}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

with

$$\Phi_{\text{ener}} \subset \Phi_{\text{dyn}}$$

(h) Quantum structures

$$(\mathcal{H}, A, \hbar) = \Phi_{\text{quant}} \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$$

6.10 Mass and Energy as Emergent Structural Properties

In the PSE, mass and energy are neither fundamental quantities nor intrinsic physical properties. They emerge as structural characteristics of the pregeometric domain $\left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)}$, and are represented

as physical quantities only through the projection Φ . Their nature is therefore ontologically pre-physical and depends exclusively on the properties of the emergent structure.

Mass as Emergent Rigidity of the Pattern

Mass is defined as the *emergent rigidity* of the correlation pattern within the pregeometric domain. It measures the amount of structural coherence required to maintain a configuration stable across emergent time. A configuration endowed with high rigidity resists changes in its emergent form, exhibiting a structural persistence that, in the physical domain, is represented as inertia.

Formally, mass is a property of the emergent triplet $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$, and cannot be defined in the fundamental domain nor in the physical domain without the intervention of the projection Φ . Its existence is therefore a direct consequence of the fundamental reorganization $\mathfrak{D}$ and of the stability of emergent patterns.

Energy as Emergent Tension of the Pattern

Energy is defined as the *emergent tension* of the pattern, namely the resistance to structural relaxation imposed by the $E + C$ principle. It measures how strongly an emergent configuration opposes the reduction of its internal complexity. A highly coherent configuration exhibits high emergent tension, which in the physical domain is represented as energy.

Energy is also a property of the non-observable emergent domain and cannot be defined at the fundamental level. Its physical representation is obtained through the map Φ_{ener} , which translates emergent tension into measurable physical quantities.

Structural Equivalence Between Mass and Energy

In the PSE, mass and energy are not distinct quantities: they are two manifestations of the same structural property, the *emergent coherence* of the pattern. Mass measures resistance to changes in the configuration; energy measures resistance to relaxation. Since both arise from the rigidity and tension of the emergent pattern, their equivalence is an ontologically inevitable consequence of the structure, not a postulated physical principle.

The physical relation between mass and energy is therefore the image, through Φ , of a deeper and pregeometric equivalence. The familiar physical relation between mass and energy is not fundamental, but represents the geometric compression of a structural property of the emergent domain.

Role of the Projection Φ

The projection Φ does not create mass and energy; it provides a physical representation consistent with the emergent geometry. In the physical domain, mass and energy appear as measurable quantities, but their origin is entirely structural. The distinction between mass and energy is therefore an artifact of the physical representation, whereas in the emergent domain they constitute complementary aspects of the same ontological property.

6.11 Other Emergent Properties of the Structure

Beyond mass and energy, the emergent domain $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ gives rise to additional properties that, in the physical domain, are represented as fundamental quantities or structures. These are not physical entities at the ontological level: they are characteristics of the form, coherence, and

symmetries of the emergent pattern, which the projection Φ translates into measurable physical quantities.

Emergent Spin

Spin is not an intrinsic property of physical objects, but an emergent symmetry of the correlation pattern. Its discrete representation in the physical domain results from the compression performed by the projection maps, which translate coherent configurations into quantized values.

Emergent Charge

Charge is a structural asymmetry of the emergent pattern. It does not exist in the fundamental domain and is not a primary physical quantity: it is a property of the emergent correlation C_e , which Φ represents as a discrete quantity associated with physical fields.

Physical Fields as Compressions

Fields are not ontological entities. They are images of coherent variations in $D_S^{(e)}$ and C_e , compressed by Φ_{field} into continuous and localized structures. Their dynamics are an effective representation of the structural reorganization of the emergent domain.

Emergent Geometry

Geometry is not assumed at the ontological level. It is a compression of the emergent structural distance $D_S^{(e)}$ through Φ_{geom} . The metric properties of the physical domain reflect limits and regularities of the emergent structure, not fundamental characteristics of space.

Emergent Dynamics

Physical dynamics are an effective representation of variations in the emergent pattern. They do not exist in the fundamental domain and cannot be assumed as primary properties. They result from the compression of structural transformations into local and temporal physical laws.

Emergent Symmetries

Physical symmetries are images of structural regularities in the emergent domain. They are not fundamental properties of physics, but consequences of the coherence of the pattern and the stability of correlations. Φ translates these regularities into geometric and gauge symmetries.

Emergent Physical Constants

Physical constants (c , $\hbar$, G) are not fundamental parameters. They are structural limits of the emergent domain, represented in the physical domain as universal numerical values. Their universality reflects the stability of the emergent structure, not an intrinsic property of physics.

Summary

All observable physical properties — spin, charge, fields, geometry, dynamics, symmetries, and constants — are effective representations of structural characteristics of the emergent domain. None of them are fundamental: they are compressed images of regularities, tensions, rigidities, and symmetries of the pattern $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$. The projection Φ does not create these properties, but translates them into physical forms consistent with the emergent geometry.

7. General properties of the emergent physical domain

The emergent physical domain is neither an extension of the fundamental domain nor a reconstruction of it. It is an effective representation of the pregeometric structure $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$ obtained through the projection map Φ . Physical properties are not primitive: they emerge as compressed images of the emergent correlation and the emergent structural distance. This section

describes the general features of the physical domain that follow inevitably from the ontological hierarchy of the PSE.

7.1 Non-fundamentality of physical quantities

In the physical domain, no quantity is fundamental. Distance, metric, time, dynamics, fields, statistical distributions, symmetries, and energy are effective representations of the emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$.

Physical constants are not primitive: they are images of structural parameters of the fundamental domain (such as the structural limit c).

The physical domain contains no original information: it is a monotonic compression of the emergent structure and cannot reconstruct the higher levels of the hierarchy.

7.2 Emergent locality

Physical locality belongs to the emergent domain and appears when the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ reaches a level of complexity capable of generating stable and coherent configurations. The projection map Φ represents this organization within the physical domain, translating structural order into geometric order.

Features of Emergent Locality

- Physical locality expresses the organization of the emergent structure.
- Its presence depends on the complexity achieved by the emergent domain.
- Its stability varies across different classes of Φ -maps and across distinct emergent configurations.
- Phenomena such as entanglement show that physical distance may diverge from structural proximity, revealing correlations that traverse the geometric ordering.

Entanglement

Entanglement arises from structural proximity within $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$: physically separated points can share strong correlations when they occupy contiguous regions in the emergent structure.

7.3 Emergent causality

Physical causality belongs to the emergent domain and appears when the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ reaches a level of complexity sufficient to define a stable causal order. The projection Φ represents, in the physical domain, the geometric and dynamical form taken by the emergent structure, making visible:

- **an effective temporal ordering,**
- **a causal structure C** consistent with the emergent organization,
- **a propagation limit,** the physical image of the structural limit c .

Physical causality is a derived phenomenon, rooted in the configuration of the emergent domain and in its representation through Φ . Its presence and specific form depend on the class of emergent structures and on their corresponding physical representations.

7.4 Physical time as an effective representation

Physical time belongs to the emergent domain and takes shape when the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ reaches a level of complexity capable of supporting a stable temporal order. The structural index t identifies the fundamental configurations of the prespatial structure; this organization becomes physical time when the projection Φ represents it in the physical domain as a temporal coordinate or an evolution parameter.

Physical time may acquire geometric or dynamical features, such as:

- a continuous temporal coordinate,
- an evolution parameter associated with dynamics,
- an effective causal structure.

Physical time thus derives from the structural organization expressed by t and from its representation through Φ , acquiring metric, continuous, or reversible properties when the complexity of the emergent domain supports them.

7.5 Physical dynamics as compression

Physical dynamics is an effective representation of coherent variations in the emergent correlation. The projection Φ expresses these variations as physical evolution, producing equations of motion that compress structural regularities into a physical form.

The equations of motion reflect:

- structural regularities of the emergent domain,
- properties specific to the generated physical representation,
- informational limits of the projection.

Physical dynamics belongs to the emergent domain and takes the form supported by the complexity of the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$.

7.6. Emergent Symmetries

Physical symmetries describe the organization of laws, the structure of interactions, and the stability of phenomena. In the PSE they belong to the emergent domain and arise from the global properties of the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$. The projection Φ represents these properties in the physical domain, formalizing them as geometric, dynamical, and discrete invariants.

Symmetries originate from the mathematical constraints of the emergent structure: continuity, isotropy, saturation, stability, and coherence. The representation Φ expresses these features in physical form, making symmetries visible as properties of the compressed image of the emergent correlation.

7.6.1 Structural origin of symmetries

The emergent structure is defined by two continuous functions:

- the emergent structural distance $D_S^{(e)}$,
- the emergent correlation C_e ,

linked by a monotonically decreasing constraint:

$$C_e(x, y) = F\left(D_S^{(e)}(x, y)\right), \quad F' < 0.$$

This relation establishes that:

- the correlation reflects the full information contained in the distance,
- the distance determines the global form of the emergent structure,
- the representation Φ inherits continuity, stability, and monotonicity.

Physical symmetries therefore express global properties of the continuous and isotropic structure

$$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}.$$

7.6.2 Lorentz symmetry as a property of the emergent metric

The continuity and regularity of the emergent structural distance are reflected in the geometric form represented by Φ . A continuous and isotropic structure without privileged directions produces a physical representation consistent with regular linear transformations.

From this organization arise:

- invariance under regular linear transformations,
- isotropy of the emergent domain,
- absence of privileged geometric scales.

Lorentz symmetry originates from the regularity and isotropy of the emergent structure and is formalized by Φ in the physical metric.

7.6.3 Gauge symmetries as a property of the emergent correlation

The emergent correlation C_e is:

- continuous,
- non-local,
- compressive,
- monotonic,
- determined by $D_S^{(e)}$.

These features imply that the correlation is defined up to transformations that preserve the functional constraint linking distance and correlation.

Gauge symmetries represent transformations that:

- preserve the emergent correlation,

- maintain the compressive character of the representation Φ ,
- leave physical quantities invariant.

The compressed correlational structure expressed by Φ formalizes gauge symmetries in the physical domain.

7.6.4 Invariance of physical laws

Emergent physical laws retain their form under transformations that preserve the emergent structure. Since Φ is monotonic and compressive, physical invariance appears with respect to:

- transformations preserving the emergent metric,
- transformations preserving the emergent correlation,
- transformations preserving compressivity.

Physical invariance thus derives from the global coherence of the emergent structure and from its representation in the physical domain.

7.6.5 Stability of symmetries

The stability of emergent physical fields reflects the global coherence of the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$. Symmetries maintain their form under local variations of the emergent structure, showing that:

- Lorentz symmetry is stable,
- gauge symmetries are stable,
- discrete symmetries are stable.

The stability of symmetries belongs to the emergent domain and arises from the continuity and coherence of the structure represented by Φ .

7.6.6 Summary

Physical symmetries arise from the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$. Φ represents them as invariants of the emergent form. Their origin is structural, their form is geometric, their stability is global.

7.7. Emergent Physical Constants

Fundamental physical constants — the correlation scale $\hbar$, the structural limit c , and the compressivity coefficient G — belong to the emergent domain. In the PSE they arise from the global properties of the pregeometric structure and from the form taken by the emergent correlation $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$. The projection Φ represents these properties in the physical domain, translating them into universal constants of effective physical theories.

Physical constants emerge from three structural elements of the fundamental domain:

- **discreteness of the prespatial domain,**
- **binary nature of the fundamental correlation,**
- **structural threshold Δ** selecting admissible correlations.

These elements define minimal scales, global limits, and response coefficients that Φ represents as physical constants.

7.7.1 Structural origin of physical constants

The ontological chain of the PSE:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} (X, D_S^{(e)}, C_e)_{\text{reg}} \xrightarrow{\Theta_k} (X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

organizes the emergent structure through three fundamental steps:

1. Discreteness of the fundamental domain

The prespatial structure is discrete and introduces:

- ontological granularity,
- a minimal structural scale,
- original non-continuity.

This granularity becomes the basis of minimal physical scales.

2. Binary correlation C_f

The fundamental correlation takes discrete values:

$$C_f(i, j) \in \{0, 1\},$$

and introduces:

- structural quantization,
- discrete selectivity,
- informational rigidity.

This selectivity is the root of physical quantization.

3. Structural threshold Δ

The threshold Δ selects the correlations that contribute to the emergent structure, producing:

- saturation,
- regularity,
- global coherence.

Δ determines which regions of the fundamental domain participate in the emergent form.

Conceptual result

These three elements generate three universal scales:

- $\hbar$ as the minimal scale of emergent correlation,
- c as the upper limit of admissible structural reorganization,
- G as the compressivity coefficient of the physical representation.

Physical constants belong to the emergent structure and are represented by Φ as universal parameters of physical theories.

7.7.2 Emergence of the $\hbar$ scale

The $\hbar$ scale expresses the minimal variation of the emergent correlation. The binarity of C_f and the selectivity imposed by Δ produce a granularity that survives collective regularization, generating a minimal variation of the correlation:

$$\hbar_{\text{structural}} = \min \Delta C_e.$$

This scale represents:

- the minimal granularity of the emergent correlation,
- the structural basis of quantization,
- the informational threshold of physical variations.

Φ represents this scale as the Planck constant in the physical domain.

7.7.3 Emergence of the structural limit c

The limit c expresses the maximal amplitude of admissible reorganizations of the emergent structure. The emergent distance $D_S^{(e)}$ possesses a global saturation that defines an upper bound for structural variations.

The representation Φ translates this bound into physical form as:

- the maximal speed of observable variations,
- the limit of the emergent metric,
- the relativistic constant of effective theories.

c belongs to the emergent structure and is represented by Φ as the physical speed limit.

7.7.4 Emergence of the coefficient G

The coefficient G expresses the response of the physical representation to the compressivity of the emergent structure. The projection Φ compresses the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ into the physical domain, producing:

- curvature as a compressed image of the emergent distance,
- mass as a compressed image of saturated regions,
- a mass–curvature relation determined by the degree of compression.

G represents the response coefficient of the physical metric to structural compression.

7.7.5 Summary

Physical constants arise from the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$. Φ represents them as universal scales of the emergent form. $\hbar$ expresses minimal granularity, c the structural limit, G the compressive response.

7.8 Non-Invertibility of the Physical Domain

The physical domain contains less information than the emergent domain. Consequently:

- no physical quantity can reconstruct $D_S^{(e)}$;
- no physical structure can reconstruct $D_S^{(f)}$;
- no physical dynamics can influence the fundamental structure.

Physics is ontologically non-retroactive. The physical domain is a compressed representation, not a fundamental level.

8. Connection with Operational Physics

Operational physics describes the physical domain through metrics, fields, dynamics, and discrete quantities. In the PSE these structures are not fundamental: they belong to the emergent domain and are represented by Φ as compressive images of the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$.

Known physical theories — general relativity, quantum mechanics, field theory, and classical dynamics — correspond to different regimes of the same emergent structure. Their mutual consistency is not imposed: it follows from the global coherence of the emergent form.

8.1 Emergence of General Relativity (GR)

General relativity treats the metric and curvature as fundamental elements. In the PSE, both metric and curvature belong to the emergent domain and are represented by Φ as geometric forms of the structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$.

Conceptual origin

The emergent distance is:

- continuous,
- regular,
- isotropic,
- saturating.

The representation Φ expresses these properties as a physical metric that is coherent and free of privileged directions.

Operational regime

GR emerges as an effective limit when:

- compressivity is weak,
- the emergent structure is nearly isotropic,
- the emergent correlation is stable.

In this regime, the metric represented by Φ reproduces:

- geodesics,

- regular curvature,
- tensorial structures,
- geometric dynamics.

General relativity is therefore an effective representation of the emergent form.

8.2 Emergence of Quantum Mechanics (QM)

Quantum mechanics is characterized by nonlocality, minimal granularity, and non-classical correlations. In the PSE these properties arise from the form taken by the emergent correlation.

Conceptual origin

The emergent correlation C_e :

- is continuous,
- is nonlocal,
- is compressive,
- derives from a binary correlation.

Collective regularization preserves a minimal variation of the correlation, defining a structural scale:

$$\hbar_{\text{structural}} = \min \Delta C_e.$$

This scale represents the minimal granularity of the emergent correlation.

Operational regime

QM emerges as an effective limit when:

- compressivity is strong,
- the emergent structure is highly correlated,
- the minimal variation of the correlation dominates the dynamics.

In this regime Φ represents:

- nonlocality,
- minimal granularity,
- action thresholds,
- state spaces consistent with the emergent structure.

8.3 Emergence of Quantum Field Theory (QFT)

Quantum field theory describes continuous fields, operators, and propagators. In the PSE, fields arise as compressive images of the emergent correlation.

Conceptual origin

The emergent correlation is:

- continuous,

- regular,
- saturating.

The compressive nature of Φ produces physical fields that are:

- continuous,
- regular,
- free of uncontrolled divergences.

Operational regime

QFT emerges as an effective limit when:

- the emergent correlation is quasi-local,
- the emergent metric is regular,
- compressivity is moderate.

In this regime the fields represented by Φ reproduce regular propagators and effective local interactions.

8.4 Emergence of Classical Dynamics

Classical dynamics describes trajectories and equations of motion. In the PSE, physical dynamics is a compressive representation of coherent variations in the emergent structure.

Conceptual origin

The emergent correlation and the emergent metric define a geometric and dynamical order that Φ expresses as physical evolution.

Operational regime

Classical dynamics emerges as an effective limit when:

- the emergent correlation is weak,
- the emergent metric is nearly flat,
- compressivity is minimal.

In this regime the equations of motion represented by Φ coincide with classical ones.

8.5 Unity of Operational Theories

Operational physical theories are not independent: they represent different ways in which the emergent structure is expressed by Φ . Their compatibility follows from the global coherence of the form $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$. Each physical theory corresponds to a specific regime of correlation, saturation, and compressivity.

8.6 Summary

The physical domain is a compressive representation of the emergent structure. Φ does not generate physics: it formalizes it as a coherent image of the pregeometric form. GR, QM, QFT, and classical

dynamics are different regimes of the same structure $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$. Their unity follows from the ontological coherence of the PSE.

9. Synthesis of the Conceptual Framework

The Conceptual Framework of the PSE establishes that fundamental reality is not physical, geometric, or dynamical, but structural. The prespatial structure S is ontologically real while remaining unobservable and lacking coordinates, metric, causality, or dynamics. The relations among the elements of S are reorganized by the fundamental reorganization $\mathfrak{D}$, which generates the discrete fundamental structural distance $D_S^{(f)}$, responsible for the internal coherence of the structure. From $D_S^{(f)}$, through the ontological threshold Δ , derives the fundamental correlation C_f , binary and selective.

The aggregation of fundamental correlations produces the continuous emergent structural distance $D_S^{(e)}$ and the emergent correlation C_e , both unobservable yet already regular. These quantities constitute the emergent domain X , a pregeometric level containing continuous relational information that is not yet physical.

Physical quantities emerge only when the emergent structure reaches a sufficient level of complexity. The monotonic projection

$$\Phi: \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{phys}}$$

compresses the pregeometric structure into effective representations: geometry, dynamics, fields, statistical distributions, discrete structures, and quantum structures. Φ does not introduce any autonomous physicality: it formalizes the portion of the emergent structure that has already acquired enough complexity, stability, and coherence to be represented as physical quantity.

Each level of the ontological chain

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} \left(X, D_S^{(e)}, C_e\right)_{\text{reg}} \xrightarrow{\Theta_k} \left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

contains less information than the previous one and is non-retroactive: no physical quantity can reconstruct or influence the fundamental structure. Observability is an emergent property that appears only in the physical domain formalized by Φ .

Fundamental reality also gives rise to emergent physical constants: the threshold Δ and structural discreteness imply a minimum correlation scale that appears in the physical domain as $\hbar$; the structural limit c becomes, in the physical representation, the relativistic bound; the compressivity of Φ produces a response coefficient that in the physical domain takes the form of the gravitational constant G . None of these constants is fundamental: all are compressed images of structural constraints of the emergent domain.

The Conceptual Framework shows that observable physics is not a set of primitive axioms, but the compressed manifestation of a real prespatial structure. Geometry, dynamics, causality, fields, and quantum correlations emerge as effective representations of structural correlation, providing a unified ontological basis for the emergence of physical quantities and for the compatibility between local and non-local phenomena.

Part III - Theoretical Positioning

The PSE arises from a methodological observation: a theory of physical emergence can be coherent only if the ontological level contains no physical concept.

From this premise follows a negative analysis: by eliminating all structures that introduce physics into the ontological level, what remains is a minimal chain of non-physical, non-geometric, non-dynamical, and non-causal elements. This chain is not a creative choice but the result of an exclusion process: what remains when everything that generates incoherence is removed.

The PSE is therefore not a speculative proposal, but an inevitable structure for anyone who wishes to describe physical emergence from a non-physical source without incurring circularity or implicit assumptions.

1. Unresolved Problems in Emergentist Physics

1.1 Geometric Circularity

Many emergentist approaches attempt to derive spacetime geometry from more primitive structures. However, these structures already incorporate implicit geometric elements: nodes with adjacencies, graphs with connections, networks with discrete distances, correlations with underlying topologies. In such cases, geometry does not emerge: it is presupposed. The theory becomes circular because what should be derived is already present at the ontological level.

1.2 Dynamical Circularity

Similarly, several models assume dynamics at the pre-physical level: state evolutions, node updates, transition rules, unitary or stochastic operators. If dynamics is already present at the ontological level, then physical dynamics does not emerge: it is inherited. This prevents explaining the origin of physical dynamics and makes a non-circular emergence theory impossible.

1.3 Failure to Produce Discrete Quantities

Discrete physical quantities (mass, spin, charge, quantized states) do not emerge from correlations or pregeometric structures unless they are explicitly introduced. Correlations alone do not select or stabilize discrete values: they produce patterns, not quantities. To obtain discrete physical quantities, a mechanism of **compression and selection** is required—absent in all known emergentist theories. This leaves unresolved one of the central problems of fundamental physics.

1.4 Incompatibility Among Emergent Physical Domains

Emergentist theories tend to generate a single domain: geometry, or dynamics, or discrete structures. There is no framework that simultaneously generates:

- geometry,
- discrete quantities,
- dynamics,
- stable configurations,

from a single source. The lack of multi-emergence leads to internal incompatibilities: geometry and particles do not derive from the same structure, and dynamics is not coherent with discrete quantities.

1.5 Absence of a Coherent Non-Physical Source

Many models assume implicit physical concepts at the ontological level: causality, nodes, states, topologies, metrics, continuity or discreteness. This prevents defining a truly non-physical source. A theory of physical emergence requires that the ontological level be free of any physical concept; otherwise, emergence becomes impossible or circular.

1.6 Dependence on Implicit Physical Structures

Even when not explicitly declared, many theories incorporate:

- temporal succession,
- evolution,
- causal relations,
- notions of distance,
- topological constraints.

If these elements are present at the ontological level, their emergence cannot be explained. A coherent theory must avoid any form of implicit physics in the pre-physical level.

2. Negative Analysis: What Cannot Work

Physical emergence requires that the ontological level be devoid of any physical concept. This principle, seemingly simple, has drastic consequences: it eliminates almost all the structures used in contemporary emergentist theories. The present analysis does not yet introduce the PSE: it shows what **cannot** constitute a coherent ontological level.

2.1 Why Networks Cannot Generate Geometry

Many pre-geometric models assume networks, graphs, or discrete structures as an ontological basis. However, such structures inevitably incorporate:

- adjacencies,
- connections,
- relations of proximity,
- topological constraints.

These elements are already forms of implicit geometry. If geometry is present at the ontological level, it cannot emerge: it is simply transferred. A theory of geometric emergence must therefore exclude any structure that contains spatial relations, even if discrete.

2.2 Why Non-Physical Correlations Cannot Generate Discrete Physical Quantities Without a Compression Map

Correlations, even when emergent, do not select or stabilize discrete values. A pregeometric correlation may contain patterns, constraints, saturations, rigidity, and compatibility structures; it may exhibit continuity, coherence, and regularity. But **it does not contain physical quantities**. Discrete quantities — mass, spin, charge, quantized states — require a mechanism of **compression and selection** that transforms emergent constraints into discrete physical values.

Consequently, no correlation structure,

$$C_f, C_e,$$

can generate a discrete physical quantity without a compression map:

$$\Phi: E_{\text{reg}}^{(k)} \longrightarrow \{\text{discrete physical quantities}\}.$$

Without such a mechanism, correlations cannot produce discrete physical values: they remain **structural behaviors**, not physical quantities.

The map Φ formalizes the level of complexity of the emergent structure from which physical quantities can **spontaneously emerge**. It does not introduce physics into the ontological domain: it extracts discrete quantities **only when** the emergent structure has reached a regime of saturation, compatibility, and rigidity sufficient to be represented as a physical value.

In other words:

- **non-physical correlations do not become physical on their own;**
- **discreteness does not arise from correlation**, but from the **monotonic compression** performed by Φ ;
- **physics is a representation of emergent behavior**, not a content of the emergent level.

2.3 Why Dynamics Cannot Be Fundamental

Any form of dynamics at the ontological level inevitably introduces:

- succession,
- change,
- evolution,
- causality,
- implicit time.

These elements are physical concepts. If dynamics is fundamental, then physical dynamics does not emerge: it is inherited from the ontological level. A theory of emergence must therefore exclude any form of ontological evolution, even when not interpreted as physical time.

In particular, no ontological rule of the form

$$X \longrightarrow X'$$

can be admitted, because it already encodes temporal succession—i.e., an implicit physical order.

2.4 Why Causality Cannot Be Assumed

Causality is a physical relation. Assuming it at the ontological level means introducing:

- directionality,
- order,
- succession,
- temporal structure.

An emergentist theory that assumes causality cannot explain the emergence of physical causality. Causality must be an emergent phenomenon, not an ontological element.

Thus, no ontological relation of the form

$$x \prec y$$

can be present at the fundamental level.

2.5 Why No Physical Structure Can Be Present at the Ontological Level

Any physical concept introduced at the ontological level—even implicitly—prevents emergence. Among these:

- distance,
- metric,
- topology,
- continuity,
- discreteness,
- states,
- nodes,
- operators,
- evolution,
- succession.

A theory of physical emergence must be built on a completely non-physical ontological level. This eliminates almost all structures used in known emergentist theories.

In particular, the ontological level cannot contain any object of the form:

$$d(x, y), g_{\mu\nu}, \mathcal{T}, \rho, \hat{H}, \hat{U}(T),$$

nor any state space, discrete or continuous.

3. The Ontological Chain of the PSE as an Inevitable Structure

The negative analysis shows that no theory can assume physical, geometric, or dynamical elements at the ontological level without generating circularity. Once every form of physical contamination is removed, only one structure remains possible: a sequence of non-physical elements which, when arranged coherently, allows physics to emerge **without presupposing it**.

This sequence is the **ontological chain of the PSE**, the only one that does not introduce physical assumptions at the fundamental level and that enables the emergence of **all observable physical domains**.

3.1 The ontological chain

The ontological chain consists of a minimal sequence of non-physical elements, each necessary to avoid circularity:

$$S \rightarrow D_S^{(f)} \rightarrow C_f \rightarrow D_S^{(e)} \rightarrow C_e \rightarrow E_{\text{reg}} \rightarrow E_{\text{reg}}^{(k)} \rightarrow \Phi \rightarrow \mathcal{D}_{\text{physical}}$$

In brief:

- S : prespatial, real, non-geometric source.
- $D_S^{(f)}$: fundamental structural distance, non-physical.
- C_f : fundamental correlation, selected through Δ .
- $D_S^{(e)}$: emergent structural distance, non-physical.
- C_e : emergent correlation, non-physical.
- $E_{\text{reg}}, E_{\text{reg}}^{(k)}$: regularized and filtered emergent structure.
- Φ : non-ontological compression into physical domains.
- $\mathcal{D}_{\text{fisico}}$: the observable domain.

The physical domain does not belong to the ontological chain: it is a compressed image of the emergent structure.

3.2 Mathematical foundation

The chain is grounded in a minimal mathematical structure:

- a prespatial set S , without metric or causality;
- a fundamental distance $D_S^{(f)}$ and a fundamental correlation C_f , selected through Δ ;
- monotonic families of operators F_λ and G_λ that generate the emergent quantities $D_S^{(e)}$ and C_e ;
- a regularization operator R that stabilizes the emergent structure;
- complexity thresholds Θ_k that filter the regularized structure;
- a compression map Φ that produces the physical domains.

These elements do not introduce any physical structure: they generate non-physical emergent constraints, which are then compressed into the observable domains.

3.3 Why Each Element Is Necessary

3.3.1 Ontological Elements

The Structure S

A non-physical source is required; otherwise physics would be fundamental and emergence impossible.

The Fundamental Structural Distance $D_S^{(f)}$

A fundamental distance is required; otherwise the source would lack any organization and could not generate correlations.

The Fundamental Correlation C_f

A fundamental correlation is required; otherwise no mechanism of emergence would exist.

The Emergent Structural Distance $D_S^{(e)}$

An emergent distance is required; otherwise C_f could not produce stable constraints or patterns.

The Emergent Correlation C_e

An emergent correlation is required; otherwise no constraints would exist that could be selected by the projection Φ .

3.3.2 Fundamental Ontological Constraints (Not Operators)

The Structural Index t

It is required to order the fundamental reorganization, ensuring coherence of configurations without introducing physical dynamics.

The Structural Principle $E + C$

It is required to stabilize the fundamental structure: it minimizes contradictions and tensions, ensuring that only compatible configurations persist.

The Structural Limit c

It is required to limit admissible variations between adjacent configurations, ensuring local coherence of the fundamental reorganization.

The Threshold Δ

It is required to select the relevant fundamental correlations; otherwise C_f would be indeterminate and no coherent structure would emerge.

3.3.3 Necessary Operators (Non-Ontological)

The operator F_λ

It aggregates the fundamental correlations and generates the emergent distance $D_S^{(e)}$. Without F_λ , the fundamental correlation C_f could not produce a coherent emergent metric.

The operator G_λ

It regularizes the emergent distance and generates the emergent correlation C_e . Without G_λ , the emergent structure would be unstable and not selectable.

The regularization operator R

It globally stabilizes the regularized emergent structure, producing

$$E_{\text{reg}} = (X, D_S^{(e)}_{\text{reg}}, C_{e \text{ reg}}).$$

Without R , the emergent structure could contain metric or correlational irregularities that cannot be removed by G_λ , preventing the definition of a continuous physical map.

The complexity thresholds Θ_k

These are monotonic, non-amplifying operators that filter the regularized emergent structure, imposing minimal requirements of topological, metric, correlational, and informational complexity. They produce

$$E_{\text{reg}}^{(k)} = \Theta_k(E_{\text{reg}}),$$

which represents the minimal level of complexity required for a physical map Φ to be well-defined, continuous, and compressive. Without Θ_k , the physical projection would not be universally definable.

The projection Φ

It compresses the emergent structure into the physical domain. Φ is not an ontological element: it formalizes

$$E_{\text{reg}}^{(k)}$$

into observable physical quantities, producing the physical domains $\mathcal{D}_{\text{fisico}}$.

3.3.4 Non-Ontological Result

The Physical Domains $\mathcal{D}_{\text{physic}}$

They are the final result: what we observe as discrete quantities, geometry, dynamics, fields, statistics, and quantum structures.

3.4 Why Nothing Can Be Removed

If any ontological element is removed:

$$\text{Chain} \setminus X \Rightarrow \text{ontological collapse.}$$

Ontological Elements

- **without** S , no ontological source exists;
- **without** $D_S^{(f)}$, S has no internal organization and cannot generate correlations;
- **without** C_f , no emergence exists;
- **without** $D_S^{(e)}$, no emergent structure exists;
- **without** C_e , no emergent constraint exists.

Fundamental Ontological Constraints (Non-Operators)

- **without** t , the fundamental reorganization cannot be ordered and the emergent structure does not stabilize;
- **without** $E + C$, no structural stabilization exists and the fundamental distance $D_S^{(f)}$ cannot emerge;
- **without** Δ , no selection of fundamental correlations exists and C_f is indeterminate;
- **without** c , no structural limit exists and the fundamental reorganization loses local coherence.

Necessary Operators (Non-Ontological)

- **without** F_λ , there is no aggregation of the fundamental correlations and $D_S^{(e)}$ cannot emerge;
- **without** G_λ , there is no regularization of the emergent distance and C_e cannot emerge;
- **without** R , the emergent structure cannot be globally stabilized, and metric or correlational irregularities would prevent the definition of a continuous physical map;
- **without the complexity thresholds** Θ_k , the regularized emergent structure cannot reach the minimal complexity required for a continuous and compressive physical projection, and Φ cannot be universally defined;
- **without** Φ , the emergent structure cannot be compressed into the physical domain, and no physical quantity can emerge.

Non-Ontological Result

- **without** $\mathcal{D}_{\text{physical}}$, the theory describes nothing observable.

3.5 Why Nothing Can Be Added

Adding elements means introducing implicit physical concepts:

- dynamics $\Rightarrow$ dynamical circularity;
- geometry $\Rightarrow$ geometrical circularity;
- causality $\Rightarrow$ physical contamination;
- states $\Rightarrow$ quantum contamination;
- nodes $\Rightarrow$ topological contamination.

Formally:

$$\text{Chain} + X_{\text{physical}} \Rightarrow \text{incoherence.}$$

Any addition makes the theory arbitrary or incoherent.

3.6 Why Nothing Can Be Modified

Modifying the chain means altering essential roles:

- changing the role of correlations $\Rightarrow$ loss of emergence;
- changing the role of distances $\Rightarrow$ loss of stability;
- changing the role of maps $\Rightarrow$ loss of multi-emergence;
- changing the order $\Rightarrow$ loss of ontological closure.

The chain is the only sequence that:

does not assume physics, geometry, dynamics, causality, or quantities

and generates everything we observe.

4. Multi-emergence as the distinctive property of the PSE

A theory of physical emergence is credible only if its non-physical source can generate **all** observable physical domains. Most emergentist approaches produce **a single domain** (geometry, or dynamics, or discrete structures), and fail to reconstruct the full spectrum of physical properties.

The PSE stands apart because its regularized emergent structure

$$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$$

is sufficiently rich to generate **all physical domains** through the classes of maps Φ . In compact form:

$$\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \{\text{Discrete structures, Geometry, Dynamics, Statistics, Fields, Quantum structure}\}.$$

4.1 Emergence of discrete structures

The map Φ_{disc} selects graphs, networks, and discrete quantities. These are not fundamental: they are emergent configurations of the regularized structure.

4.2 Emergence of geometry

The map Φ_{geom} compresses the emergent structure into distance, metric, and causal relations. Geometry is not ontological: it is an effective representation.

4.3 Emergence of dynamics

The map Φ_{dyn} produces an effective Hamiltonian, unitary evolution, and physical state. Time is physical, not fundamental.

4.4 Emergence of statistical properties

The map Φ_{stat} extracts distributions, entropies, and expectation values. Statistics is an informational reading of the emergent structure.

4.5 Emergence of physical fields

The map Φ_{field} generates scalar fields, vector fields, and field tensors. Fields are effective representations, not fundamental entities.

4.6 Emergence of quantum structure

The map Φ_{quant} produces the state space, continuous action, and the minimal threshold $\hbar$. Quantization is an emergent effect, not an ontological feature.

4.7 Internal compatibility between domains

Since all maps Φ_i share the same domain, the various aspects of physics emerge from **a single source**. This ensures coherence between:

- geometry and dynamics,
- fields and discrete quantities,
- statistics and configurations,
- quantum structure and classical domains.

In summary: **all physical domains are images of the same emergent structure. This multi-emergence is the unique property of the PSE.**

5. Structural Simplicity: The PSE as a Minimal Theory

The ontological chain of the PSE:

$$S \xrightarrow{\mathfrak{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} (X, D_S^{(e)}, C_e)_{\text{reg}} \xrightarrow{\Theta_k} (X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

is not an arbitrary construction. It is the **minimal structure** that allows physical emergence without introducing circularity, physical contamination, or implicit assumptions.

In this section we show that:

1. the PSE is the simplest possible theory;
2. any simpler theory is incoherent;
3. any more complex theory is arbitrary;

4. the PSE does not assume any physical concept at the ontological level;
5. the PSE does not assume geometry, dynamics, causality, topology, or discrete quantities.

5.1 Why the PSE Is the Simplest Possible Theory

The simplicity of the PSE derives from a methodological principle:

No physical concept can be present at the ontological level.

This eliminates:

- space,
- time,
- causality,
- dynamics,
- states,
- nodes,
- metrics,
- topologies,
- discrete quantities.

Once all these elements are removed, only one structure remains: **the ontological chain of the PSE.**

There is no shorter chain that is coherent.

5.2 Why Any Simpler Theory Is Incoherent

If one attempts to simplify the chain by removing any of its elements, the result is:

$\text{Chain} \setminus X \Rightarrow \text{ontological collapse or impossibility of physical emergence.}$

Examples:

- Removing $D_S^{(f)} \rightarrow$ the fundamental structure S cannot generate correlations.
- Removing $C_f \rightarrow$ emergence does not exist.
- Removing $D_S^{(e)} \rightarrow$ no stable constraints exist.
- Removing $C_e \rightarrow$ no selectable constraints exist.
- Removing $R \rightarrow$ the emergent structure cannot be stabilized, and no continuous physical map can be defined.
- Removing the complexity thresholds $\Theta_k \rightarrow$ the emergent structure does not reach the minimal complexity required, and Φ cannot be defined.
- Removing $\Phi \rightarrow$ physics does not exist.

5.3 Why Any More Complex Theory Is Arbitrary

Adding elements to the chain means introducing implicit physical concepts:

- adding dynamics → introduces time and causality;
- adding geometry → introduces distance and metric;
- adding nodes → introduces topology;
- adding states → introduces quantum structure;
- adding operators → introduces physical dynamics.

Formally:

$$\text{Chain} + X_{\text{physical}} \Rightarrow \text{incoherence.}$$

Any addition makes the theory:

- arbitrary,
- non-minimal,
- unnecessary,
- non-assessable.

The PSE is the only theory that contains nothing superfluous.

5.4 Why the PSE Assumes Nothing Physical

The ontological level of the PSE is composed of:

$$S, D_S^{(f)}, C_f, D_S^{(e)}, C_e,$$

all elements that are:

- non-physical,
- non-geometric,
- non-dynamical,
- non-causal,
- non-topological,
- non-discrete,
- non-continuous.

This guarantees that:

Physics emerges; it is not assumed.

5.5 Why the PSE Does Not Assume Geometry, Dynamics, Causality, or Topology

Geometry emerges through:

$$\Phi_{\text{geom}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow (d, g_{\mu\nu}, \mathcal{C}),$$

and is not present at the ontological level.

Dynamics emerges through:

$$\Phi_{\text{dyn}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow (\hat{H}, \hat{U}(T), \rho),$$

and is not present at the ontological level.

Causality emerges through:

$$\Phi_{\text{geom}} \rightarrow \mathcal{C},$$

and is not present at the ontological level.

Topology emerges through:

$$\Phi_{\text{disc}}: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow (G, \Gamma),$$

and is not present at the ontological level.

In summary:

No physical structure is fundamental.

6. Technical Comparison with Alternatives

This section does not aim to evaluate or criticize existing emergentist theories, but to show—technically and without polemics—why none of them resolves the structural problems identified in Section 2. The comparison is based exclusively on the criteria of:

- geometrical circularity,
- dynamical circularity,
- absence of emergent discrete quantities,
- incompatibility among domains,
- presence of physical concepts at the ontological level.

The PSE does not arise as an alternative: it arises as **what remains** once all forms of incoherence are eliminated.

6.1 Causal Set Theory

Causal Set Theory assumes as fundamental:

- discrete elements,
- causal relations,
- partial order.

Formally, the ontological level contains:

$$\{x_i\}, x_i < x_j.$$

Problems relative to the PSE criteria

1. **Fundamental causality.** The relation $<$ is an implicit physical structure. Causality cannot emerge if it is assumed at the ontological level.

2. **Implicit geometry.** The causal order determines the Lorentzian metric (Malament's theorem). Geometry does not emerge: it is presupposed.
3. **Absence of multi-emergence.** The theory generates causality and discrete structure, but not physical discrete quantities, fields, dynamics, or statistics.

6.2 Spin Networks / Loop Quantum Gravity

The ontological level contains:

$$\Gamma, j_e, i_v,$$

where Γ is a graph, j_e are spins, and i_v are intertwiners.

Problems relative to the PSE criteria

1. **Fundamental discrete structure.** The graph Γ is an implicit geometric structure (adjacencies, nodes, connections).
2. **Fundamental discrete quantities.** The spins j_e are physical quantities assumed at the ontological level.
3. **Presupposed geometry.** Areas and volumes emerge from operators built on Γ . Geometry does not emerge from a non-physical source.
4. **Absence of multi-emergence.** The theory produces quantized geometry, but not emergent dynamics, emergent fields, or emergent statistics.

6.3 Tensor Networks / MERA / PEPS

The ontological level contains:

$$\{T_i\}, \text{ connections among tensors.}$$

Problems relative to the PSE criteria

1. **Implicit lattice structure.** The tensor network is a discrete geometric structure.
2. **Fundamental operators.** Tensors are physical operators, not non-physical elements.
3. **Implicit dynamics.** Networks are constructed to represent already dynamical physical states.
4. **Absence of multi-emergence.** Networks generate states and correlations, but not physical discrete quantities, fields, or non-presupposed emergent geometry.

6.4 Emergent Gravity (Jacobson, Verlinde, Padmanabhan)

The ontological level contains:

- entropy,
- information,
- thermodynamics,
- microscopic degrees of freedom.

Problems relative to the PSE criteria

1. **Fundamental physical concepts.** Entropy, temperature, and energy are physical concepts assumed at the ontological level.
2. **Implicit geometry.** The metric is reconstructed from thermodynamic relations that presuppose causality and horizons.
3. **Absence of multi-emergence.** The theory generates geometry, but not discrete quantities, fields, or emergent dynamics.

6.5 Informational Models (Wheeler, Lloyd, Relational Rovelli)

The ontological level contains:

- bits,
- qubits,
- operators,
- states,
- informational relations.

Problems relative to the PSE criteria

1. **Fundamental operators and states.** Operators encode implicit physical dynamics.
2. **Presupposed discrete or quantum structure.** Bits and qubits are physical discrete quantities.
3. **Absence of multi-emergence.** The theory generates information, but not non-presupposed emergent geometry, nor fields, nor emergent dynamics.

7. Synthesis of the Theoretical Positioning

The Theoretical Positioning of the PSE shows that a theory of physical emergence can be coherent only if the ontological level is completely free of physical concepts. The analysis of contemporary emergentist models highlights five structural obstacles: geometrical circularity, dynamical circularity, lack of emergent discrete quantities, incompatibility among emergent domains, and the presence of implicit physical concepts in the pre-physical level. These problems are not technical details: they prevent the construction of a complete framework for physical emergence.

The negative analysis of the PSE eliminates every structure that introduces physics at the ontological level. Once every form of node, state, metric, topology, causality, succession, dynamics, or discrete quantity is removed, only one chain remains: a minimal sequence of non-physical, non-geometric, non-dynamical, and non-causal elements:

$$S \xrightarrow{\mathcal{D}} D_S^{(f)} \xrightarrow{\Delta} C_f \xrightarrow{F_\lambda} D_S^{(e)} \xrightarrow{G_\lambda} C_e \xrightarrow{R} (X, D_S^{(e)}, C_e)_{\text{reg}} \xrightarrow{\Theta_k} (X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)} \xrightarrow{\Phi} \mathcal{D}_{\text{physical}}$$

Each element is necessary:

- without S , no source exists;
- without $D_S^{(f)}$, no organization exists;
- without C_f , no emergence exists;

- without $D_S^{(e)}$, no emergent continuity exists;
- without C_e , no selectable constraints exist;
- without Φ , no physics exists;

Likewise, nothing can be added: any addition reintroduces implicit physics and generates incoherence.

The fundamental reorganization $\mathfrak{D}$, ordered by the structural index t and constrained by the limit c , maintains coherence without introducing physical time, causality, or propagation. The structural principle $E + C$ stabilizes configurations without resorting to energy, equations of motion, or physical dynamics. From this reorganization emerge the continuous distance $D_S^{(e)}$ and the emergent correlation C_e , both non-observable but already regular.

The Φ maps are not ontological objects: they are compression relations that transform non-physical emergent constraints into coherent physical domains. From them, geometry, dynamics, fields, discrete quantities, statistical distributions, and stable configurations emerge simultaneously:

$$\Phi_\alpha: \left(X, D_S^{(e)}, C_e \right)_{\text{reg}}^{(k)} \rightarrow \mathcal{D}_{\text{physical}}^{(\alpha)}.$$

This multi-emergence is a unique property of the PSE: all physical domains derive from the same non-physical source, guaranteeing internal compatibility among geometry, dynamics, fields, discrete quantities, and statistics.

The structural simplicity of the PSE is a logical consequence: it is the simplest possible structure that does not introduce physical concepts at the ontological level. Any simpler theory is incoherent; any more complex theory is arbitrary. The PSE does not arise as an alternative: it arises as **what remains** once all forms of incoherence are eliminated.

In summary, the Theoretical Positioning shows that:

- the ontological chain is inevitable;
- multi-emergence is guaranteed;
- the structure is closed, minimal, and non-retroactive;
- no physical quantity is fundamental;
- observable physics is a compressed representation of emergent correlation;
- the PSE is scientifically assessable because it eliminates all forms of circularity and physical contamination.

The PSE is not a speculative proposal: it is the minimal form of ontological coherence required to describe physical emergence from a non-physical source.

Part IV - Verifiability

The verifiability of the PSE is grounded in the emergent nature of the physical domain. The structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ produces metrics, fields, and dynamics as compressive representations through Φ . These representations possess observable forms that can be compared with experiments and empirical data.

The verifiability of the PSE is articulated along two dimensions:

- **emergent signatures**, which define specific observables;
- **anomalous data**, already detected, consistent with the emergent structure.

1 Emergent signatures and verifiable experiments

Emergent signatures are structural properties of the emergent domain that Φ expresses in the physical domain. They belong to three categories: geometric, correlative, and dynamic.

1.1 Geometric signature — Emergent metric saturation

The emergent distance is continuous, regular, and saturating. The representation Φ expresses this saturation as an upper bound on metric variations.

Emergent form High-frequency gravitational waves exhibit metric compression relative to the predictions of general relativity.

Observable Systematic deviation in the waveform at frequencies above the kilohertz regime.

Relevant experiments LIGO, Virgo, KAGRA, Einstein Telescope, Cosmic Explorer.

1.2 Quantum signature — Minimal granularity of emergent correlation

The emergent correlation possesses a minimal variation $\min \Delta C_e$. In the physical domain this variation appears as an action threshold.

Emergent form Ultra-low-energy quantum systems exhibit residual nonlinearity in superposition.

Observable Microscopic deviations from linearity in ultra-sensitive atomic interferometers.

Relevant experiments Atomic interferometry, Bose–Einstein condensates, MAQRO.

1.3 Correlative signature — Regular emergent nonlocality

The emergent correlation is nonlocal, continuous, and monotonic. The representation Φ expresses this form as regular nonlocal correlation.

Emergent form The decay of nonlocal correlation follows a monotonic regular form, distinct from that predicted by standard quantum mechanics.

Observable Systematic deviations in Bell tests at distances exceeding planetary scale.

Relevant experiments Micius satellite, terrestrial quantum networks, global-scale entanglement tests.

1.4 Dynamic signature — Emergent stability of fields

Emergent fields are saturating and free from uncontrolled divergences. Compressivity eliminates ultraviolet and infrared instabilities.

Emergent form In extreme regimes, fields exhibit stability superior to the predictions of field theory.

Observable Deviations in energy distributions in ultra-high-energy collisions or ultra-hot plasmas.

Relevant experiments

LHC, RHIC, Future Circular Collider.

1.5 Unity of emergent signatures

Geometric, correlative, and dynamic signatures are not independent: they represent different aspects of the same emergent structure $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$. Their coherence is a consequence of the pregeometric form and of the compressive nature of the Φ representation.

2 Anomalous observational data consistent with the PSE

This section does not attribute the explanation of observational phenomena to the PSE: it shows that **the emergent form of the PSE is compatible** with anomalies already detected across different areas of physics.

2.1 Cosmic acceleration

Anomaly The accelerated expansion of the universe is not explained by matter or radiation and requires the introduction of dark energy.

Consistency with the PSE The saturation of emergent distance produces a **natural geometric expansion**, represented by Φ as coherent metric dilation.

Emergent form The expansion is not caused by a physical source: it is a property of the saturating emergent structure.

Observables consistent with the PSE

- regular cosmic acceleration,
- absence of additional energetic sources,
- metrically coherent behaviour on cosmological scales.

Relevant experiments and observations

- Type Ia supernovae
- BAO (Baryon Acoustic Oscillations)
- Planck, WMAP
- Euclid, LSST

2.2 Hubble tensions

Anomaly Local and cosmological measurements of the Hubble constant do not coincide, producing a stable and unresolved tension.

Consistency with the PSE The compressivity of the Φ representation can generate **slightly different metric scales** in domains with different emergent correlation.

Emergent form The emergent structure is not perfectly homogeneous: regions with different correlation produce metric representations with slightly varying scales.

Observables consistent with the PSE

- systematic differences between local and global measurements,
- metric variations coherent with non-uniform emergent domains.

Relevant experiments and observations

- SH0ES
- Planck
- HST, JWST
- DESI

2.3 Anomalous quantum oscillations

Anomaly Ultra-sensitive quantum systems exhibit microscopic deviations from linear superposition.

Consistency with the PSE The minimal granularity $\min \Delta C_e$ produces **residual nonlinearities** at ultra-low energy.

Emergent form The emergent correlation possesses a minimal threshold that Φ represents as discrete variation of action.

Observables consistent with the PSE

- oscillations not perfectly linear,
- regular deviations in atomic interferometers,
- threshold effects in low-energy systems.

Relevant experiments and observations

- atomic interferometry (Stanford, MIT, LENS)
- Bose–Einstein condensates
- macroscopic superposition tests (MAQRO)

2.4 Anomalous stability of high-energy plasmas

Anomaly Ultra-hot plasmas exhibit stability superior to field-theoretic predictions.

Consistency with the PSE Saturation and compressivity eliminate UV divergences, producing **more stable emergent fields**.

Emergent form The Φ representation compresses extreme variations, generating regular fields even in high-energy regimes.

Observables consistent with the PSE

- plasma stability beyond QFT predictions,

- regular energy distributions,
- absence of uncontrolled divergences.

Relevant experiments and observations

- LHC Run 3
- RHIC
- Future Circular Collider

2.5 Quantum nonlocality on planetary scale

Anomaly Quantum correlations over distances > 1000 km exhibit extremely regular and stable behaviour.

Consistency with the PSE The emergent correlation is **nonlocal and monotonic**, with regular form over large distances.

Emergent form Nonlocality is not a pointwise phenomenon: it is a global property of the emergent correlation.

Observables consistent with the PSE

- regular nonlocal correlations,
- stability of correlation over planetary distances,
- coherent monotonic decay.

Relevant experiments and observations

- Micius satellite
- terrestrial quantum networks
- global-scale Bell tests

2.6 Anomalous distribution of dark matter

Anomaly Galactic rotation curves, gravitational lensing, and mass distributions do not match visible matter. No experimental evidence for dark-matter particles exists despite decades of searches.

Consistency with the PSE The emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ contains regions with **non-uniform structural rigidity**. The Φ representation expresses this rigidity as:

- coherent metric variations,
- gravitational effects without additional mass,
- emergent distributions that mimic dark-matter halos.

The emergent correlation produces **shape gradients** that appear in the physical domain as invisible mass distributions.

Emergent form Dark matter is not a particle or an additional field: it is an **elastic property of the emergent structure**, represented by Φ as a geometric contribution to the physical metric.

Observables consistent with the PSE

- regular galactic rotation curves,
- gravitational lensing without luminous sources,
- non-Gaussian mass distributions,
- halos with cored rather than cuspy profiles.

Relevant experiments and observations

- Gaia
- JWST
- DES, LSST, Euclid
- Bullet Cluster and analogous systems

2.7 Baryon Asymmetry

Anomaly The universe contains significantly more matter than antimatter, despite physical laws being largely symmetric under matter–antimatter exchange. No experimental evidence exists for mechanisms capable of producing the observed imbalance.

Consistency with the PSE The emergent structure $(X, D_S^{(e)}, C_e)_{\text{reg}}^{(k)}$ can produce **asymmetric correlation domains**, where the representation Φ expresses correlation gradients as effective matter–antimatter imbalance.

Emergent form Baryon asymmetry is not a fundamental violation: it is a **structural asymmetry** of the emergent correlation.

Observables consistent with the PSE

- absence of large-scale antimatter domains,
- uniform matter dominance across cosmological scales,
- correlation patterns consistent with asymmetric emergent domains.

Relevant experiments and observations

- CMB anisotropies (Planck)
- BBN (Big Bang Nucleosynthesis)
- LHCb (CP violation studies)
- Belle II

2.8 Macroscopic quantum coherence

Anomaly Superconductors, superfluids, and Bose–Einstein condensates exhibit coherent quantum behaviour over macroscopic distances, far beyond typical quantum-mechanical correlation lengths.

Consistency with the PSE The emergent correlation C_e is **continuous and monotonic**, allowing coherent behaviour over extended domains. Φ represents this coherence as macroscopic quantum order.

Emergent form Macroscopic coherence is a **natural property of the emergent correlation**, not an exceptional quantum phenomenon.

Observables consistent with the PSE

- long-range phase coherence,
- stable macroscopic superposition,
- regular correlation decay in extended quantum systems.

Relevant experiments and observations

- superconductivity and superfluidity
- BEC coherence measurements
- SQUID interferometry
- macroscopic entanglement experiments

2.9 Large-scale cosmic regularity (> 1 Gpc)

Anomaly The distribution of matter on gigaparsec scales is more regular and coherent than predicted by Λ CDM, with structures such as filaments and walls showing unexpected uniformity.

Consistency with the PSE The emergent correlation is **monotonic and continuous**, producing large-scale regularity in the representation Φ .

Emergent form Cosmic regularity is a **geometric expression of the emergent correlation**, not a statistical anomaly.

Observables consistent with the PSE

- coherent filaments and walls,
- regularity of large-scale structures,
- deviations from purely stochastic clustering.

Relevant experiments and observations

- SDSS
- DESI
- Euclid
- LSST

2.10 Anomalies in strong gravitational lensing

Anomaly Several strong-lensing systems show deviations from GR+ Λ CDM predictions, including anomalous magnification ratios and unexpected mass distributions.

Consistency with the PSE Regions of **non-uniform structural rigidity** in the emergent domain produce metric variations that Φ represents as lensing anomalies.

Emergent form Strong-lensing deviations are **elastic effects of the emergent structure**, not evidence of exotic matter.

Observables consistent with the PSE

- anomalous flux ratios,
- lensing without sufficient visible mass,
- coherent distortions in cluster lenses.

Relevant experiments and observations

- JWST
- HST strong-lensing surveys
- Euclid lensing catalogue
- SLACS

2.11 Low-energy neutrino oscillation anomalies

Anomaly Some low-energy neutrino experiments show oscillation patterns that deviate slightly from standard three-flavour predictions.

Consistency with the PSE The minimal granularity $\min \Delta C_e$ introduces **threshold effects** that Φ represents as small deviations in oscillation behaviour.

Emergent form Neutrino oscillations reflect **granular structure in the emergent correlation**, not new neutrino species.

Observables consistent with the PSE

- non-linear oscillation phases at low energy,
- threshold-like behaviour,
- regular deviations from standard mixing angles.

Relevant experiments and observations

- Borexino
- SNO+
- IceCube low-energy data
- JUNO (future)

2.12 Uniformity of the Cosmic Microwave Background (CMB)

Anomaly The CMB exhibits an extremely high level of uniformity between regions that, according to standard cosmological geometry, could not have been in causal contact before the emission of the radiation. This homogeneity requires the introduction of cosmic inflation as an additional dynamical phase.

Consistency with the PSE In the PSE, the emergent structure $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$ contains global pregeometric correlations. The compressive nature of the representation Φ produces metric domains that are already coherent on large scales, without requiring an inflationary dynamical mechanism. The

uniformity of the CMB is therefore compatible with the presence of global emergent correlations preceding physical geometry.

Emergent form The homogeneity does not arise from a physical process of accelerated expansion, but from the global nature of the emergent correlations that Φ compresses into a uniform metric.

Observables consistent with the PSE

- large-scale isotropy of the CMB,
- temperature coherence between regions not causally connected,
- stability of primordial anisotropies,
- no need for an inflaton field or a dedicated potential.

Relevant experiments and observations

- COBE
- WMAP
- Planck
- future CMB-S4 missions

2.13 Absence of observational data contradicting the PSE

To date, no observational data are known that require a physical structure incompatible with the emergent form $\left(X, D_S^{(e)}, C_e\right)_{\text{reg}}^{(k)}$ or with the compressive nature of the representation Φ . This absence of negative data does not imply that the PSE is correct, but it indicates that the emergent framework remains empirically compatible with the available measurements.

Part V - Ontological Vision and Perspectives

The Structural Principle of Entanglement (PSE) presents a complete and self-sufficient ontological framework in which fundamental reality is described by a discrete prespatial structure, devoid of geometry, causality, and physical quantities. From this structure emerge the continuous structural distance, the emergent correlation, and (through the monotonic and non-invertible projection Φ) all observable physical quantities.

The theory is coherent, non-retroactive, and non-invertible: each level contains less information than the previous one, and no physical quantity can reconstruct the fundamental structure. Physics is not assumed: it is a compressed representation of emergent correlation, not an ontological property of reality.

1. Closure of the Conceptual Framework

The conceptual framework of the PSE is complete. The following have been defined:

Ontological Elements

- the prespatial structure S ;
- the fundamental structural distance $D_S^{(f)}$;
- the fundamental correlation C_f ;
- the emergent distance $D_S^{(e)}$;
- the emergent correlation C_e .

Fundamental Ontological Constraints

- the structural index t (discrete ordering of reorganization);
- the structural principle $E + C$ (stabilization of configurations);
- the structural limit c (local coherence of reorganization).
- the threshold Δ (selection of fundamental correlations);

Necessary Operators (Non-Ontological)

- the aggregation and regularization operators F_λ and G_λ ;
- the regularization operator R ;
- the complexity thresholds Θ_k ;
- the compressive projection Φ

Non-Ontological Result

- the observable physical domain $\mathcal{D}_{\text{physical}}$.

Epistemic Aspects

- the general properties of the physical domain;
- indirect verifiability.

2. Role of Mathematical Formalization

The mathematical formalization of the PSE is the natural continuation of the conceptual framework. It allows one to:

- rigorously define the fundamental discrete structure;
- characterize the threshold Δ and the selectivity of correlations;
- construct the emergent structural distance $D_S^{(e)}$ through F_λ ;
- construct the emergent correlation C_e through G_λ ;
- define the monotonic (non-physical) function F ;
- formalize the regularization operator R ;
- formalize the complexity thresholds Θ_k ;
- formalize the properties of the compressive projection Φ ;
- prove theorems on the ontological hierarchy and on non-invertibility.

Mathematics does not introduce new concepts: it implements those already defined in the ontological framework.

3. General Ontological Vision

The PSE assumes as its ontological basis the prespatial structure S , composed of pure relational entities, devoid of any physical concept, connected by binary correlations and reorganized by the fundamental process $\mathfrak{D}$, stabilized by the structural principle $E + C$.

S is:

- a mathematical point without extension,
- the whole, because no “outside” exists,
- the nothing, because it contains no physical concept,
- the only possible ontological domain.

The increase in complexity of S generates a sequence of emergent levels: pure correlations, emergent coherence, emergent structure, collective rigidity, geometric compression, physical metric, physical dynamics, matter, chemistry, biology, consciousness.

Physics is not the foundation of existence: it is an emergent level of the complexity of S .

4. S as Ontological Source of Emergent Domains

S contains the ontological possibility of multiple emergences: stable, metastable, unstable, ephemeral, long-lasting, self-collapsing, and self-expanding domains.

Each emergent domain may possess:

- different constants,
- different geometries,

- different metrics,
- different dynamics,
- different symmetry breakings,
- possibility or impossibility of chemistry and biology.

These domains are not regions of space: they are emergent configurations of the complexity of S , not physical locations.

5. Physical and Conceptual Consequences

The PSE offers new interpretative keys for physical phenomena:

- entanglement as a trace of primordial nonlocality;
- indeterminacy as a property of geometric compression;
- wave–particle duality as geometric selection of emergent patterns;
- uniformity of the CMB as a consequence of the prespatial phase;
- dark matter as non-localized rigidity;
- dark energy as emergent geometric tension;
- consciousness as a non-local emergent pattern.

6. Future Directions

The main directions of development include:

- advanced metric extensions;
- emergent dynamical structures;
- emergent fields and interactions;
- correlational phenomenology;
- discrete simulations of the fundamental domain;
- experimental verifiability through strongly correlated quantum states;
- holographic and informational applications.

7. Epistemic Value of the PSE

The PSE represents:

- a unique ontological foundation,
- a natural emergent structure,
- a conceptual reconciliation between quantum and relativistic phenomena,
- an epistemic simplification of modern physics.

The PSE does not replace physics: it clarifies its place in the order of existence.

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